NON-WSQ COURSE • 1 DAY

CYBER SECURITY AWARENESS

Workshop

Recognise threats. Reduce exposure. Escalate with confidence.

C506



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Welcome to the course

Recognise

Spot threat patterns, vulnerable conditions and signs that deserve attention.

Reduce

Apply layered safeguards that protect people, devices, data and business processes.

Escalate

Communicate evidence, impact and urgency to the right person without making the incident worse.

1 DAY • 7.5 HOURS

The day at a glance

1 **Threats**

Language, attack chains and warning signs

2 **Exposure**

Assets, vulnerabilities and business risk

3 **Safeguards**

Identity, devices, data, network and people

4 **Response**

Evidence, containment and escalation

5 **Action**

Key takeaways and a 30-day action plan

Concepts on slides • detailed procedures in the Learner Guide

A safe learning environment

DO

- Use only supplied mock data
- Stay inside named training sites
- Preserve evidence and privacy
- Stop when scope is uncertain

DO NOT

- Test real organisations
- Enter real passwords or personal data
- Attempt denial of service
- Publish exploitable findings

Cybersecurity is a business capability

Confidentiality

Only authorised people and systems access information.

Integrity

Information and processes remain accurate, complete and trustworthy.

Availability

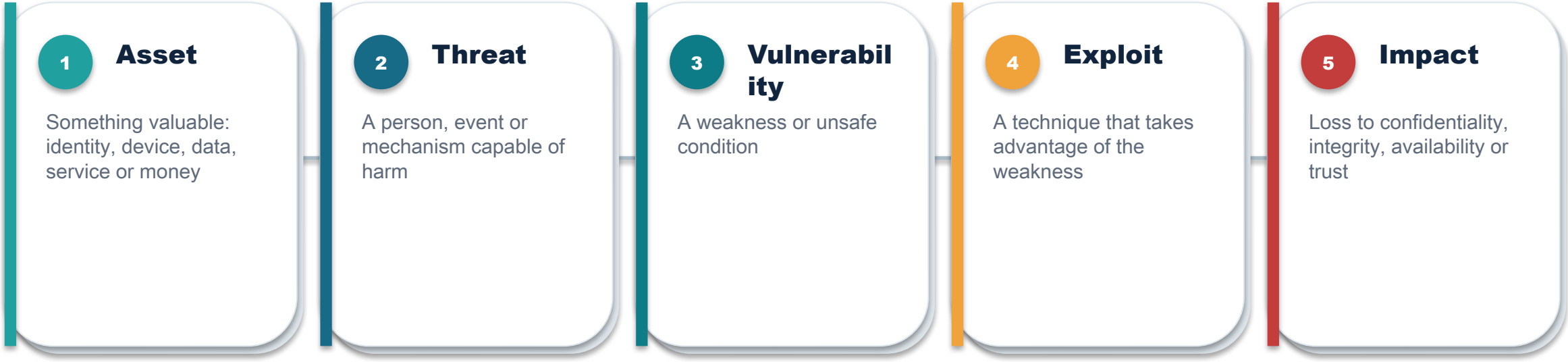
Services and data are accessible when the business needs them.

Safety & trust

Digital failures can affect people, operations, money and reputation.

Source: https://www.ibm.com/think/topics/cybersecurity

How a cyber incident unfolds



Risk grows when valuable assets, credible threats and exploitable weaknesses meet.

Source: https://www.ncsc.gov.uk/collection/vulnerability-management/understanding-vulnerabilities

RISK THINKING

Risk is a decision, not just a score

Likelihood

How plausible is exploitation in this context? Consider reachability, capability and active exploitation.

Impact

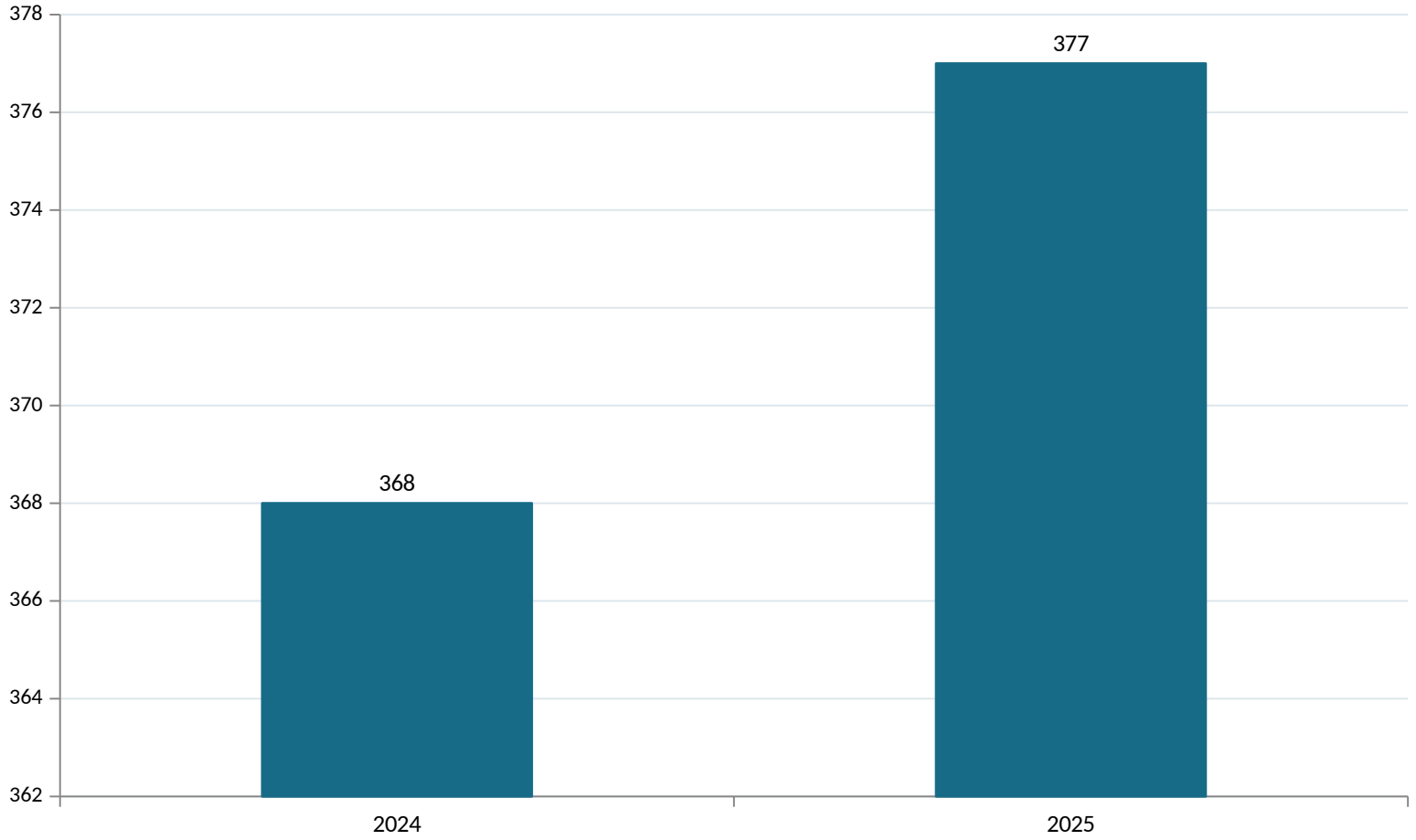
What happens to people, operations, finance, compliance and reputation?

Control strength

Does a safeguard prevent, detect, contain or recover — and has it been verified?

EDITABLE DATA VIEW

Business email compromise cases edged upward



READ THE SIGNAL

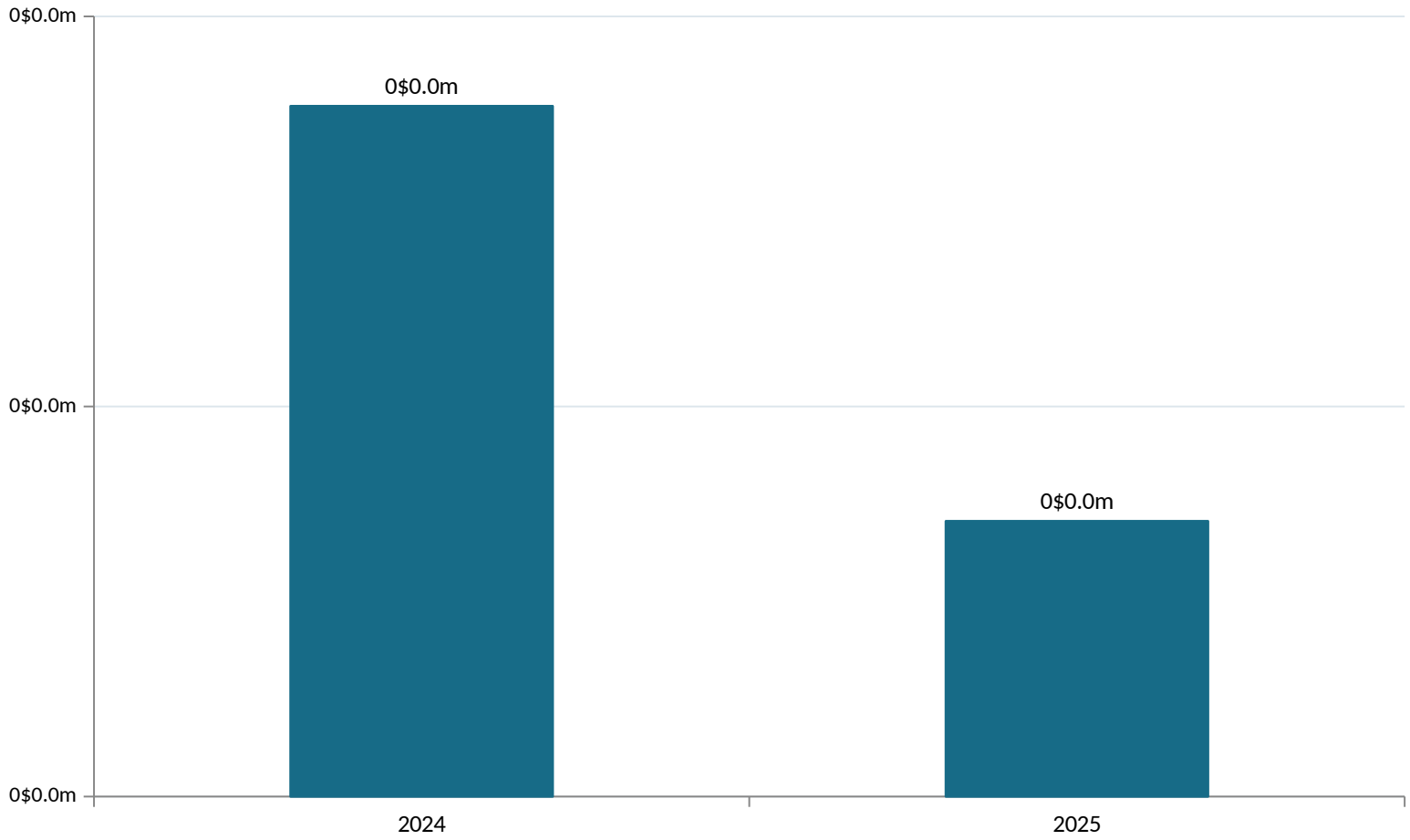
Singapore Police recorded 377 BEC cases in 2025, up from 368 in 2024. Small changes in volume still represent repeated opportunities for high-impact loss.

Chart is native and editable in PowerPoint.

Source: https://www.police.gov.sg/-/media/SPF/Media-Room/Statistics/Annual-Scams-and-Cybercrime-Brief-2025/Ann…

EDITABLE DATA VIEW

BEC losses fell — but remained material



READ THE SIGNAL

Reported losses fell, yet S$35.3 million remained at stake. Independent verification and rapid escalation change outcomes.

Chart is native and editable in PowerPoint.

Source: https://www.police.gov.sg/-/media/SPF/Media-Room/Statistics/Annual-Scams-and-Cybercrime-Brief-2025/Ann…

Personal and business risk are connected

Personal account

A reused password unlocks email; email resets other accounts.

Work identity

A personal device or inbox becomes a path into business systems.

Business process

A believable message changes payment, access or recovery decisions.

Shared consequence

Money, data, services and trust are affected across both worlds.

01

TOPIC

Identify cyber risks and threats

Read behaviours, attack paths and warning signs before deciding what to do.

SEE

DECIDE

ACT

Threat actors pursue different goals

Cybercriminal

Money through theft, extortion, fraud or resale.

Insider

Malicious, coerced or accidental misuse of legitimate access.

Hacktivist

Disruption or disclosure to advance a cause.

Nation-state / APT

Strategic intelligence, influence or critical-infrastructure access.

Opportunist

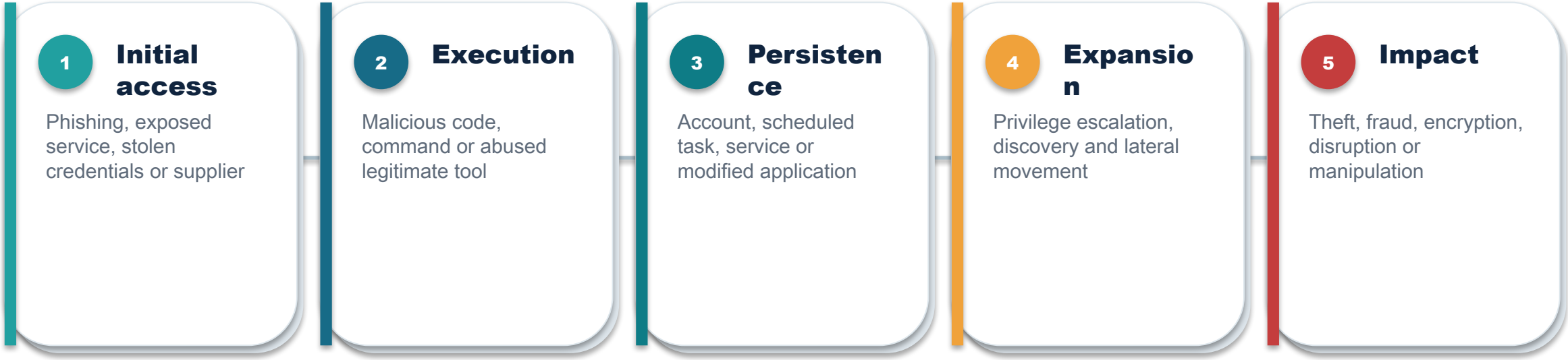
Scans widely for exposed or unpatched systems.

Supplier compromise

Abuses trusted software, services or relationships.

THINK IN SEQUENCES

Attack chain: from initial access to impact



Break one link early and the whole outcome can change.

Malware families: behaviour matters

Virus

Attaches to a host file and executes when the file runs

Worm

Self-propagates across reachable systems

Trojan

Disguises malicious capability as legitimate software

Ransomware

Encrypts or disrupts systems and demands payment

Spyware

Collects activity, credentials or data covertly

Rootkit

Hides privileged malicious activity

Bot

Places a device under remote command

Logic bomb

Triggers a malicious action when a condition is met

Adware

Displays intrusive advertising and may track browsing or alter settings

Keylogger

Records keystrokes to capture credentials and sensitive input

Fileless malware

Abuses memory and legitimate administration tools instead of relying on a normal file

Cryptojacker

Hijacks computing resources to mine cryptocurrency

Source: https://www.crowdstrike.com/en-us/cybersecurity-101/malware/types-of-malware/

OBSERVABLE SIGNALS

Malware warning signs

Virus

Unexpected file changes; antivirus detections

Worm

Rapid scanning and repeated connections

Trojan

Untrusted installer, new process or persistence

Ransomware

File extensions change, ransom note, unavailable services

Spyware

Browser changes, unusual outbound traffic

Rootkit

Integrity anomalies, unexplained privileged changes

Bot

Beaconing, DDoS or spam activity

Logic bomb

Scheduled or conditional destructive code

Adware

Persistent pop-ups, redirects and unwanted extensions

Keylogger

Unexplained input hooks, suspicious processes or credential theft

Fileless malware

Unexpected PowerShell, script hosts, encoded commands or memory activity

Cryptojacker

High CPU or cloud cost, overheating and unknown mining connections

Source: https://www.crowdstrike.com/en-us/cybersecurity-101/malware/types-of-malware/

Virus, worm and trojan are not interchangeable

PROPAGATION

- Virus needs a host and user execution
- Worm self-propagates across reachable systems
- Trojan relies on disguise and user trust
- All may deliver additional payloads

DEFENSIVE LENS

- Control execution
- Patch and segment
- Use trusted software sources
- Monitor behaviour, not only filenames

Source: https://www.crowdstrike.com/en-us/cybersecurity-101/malware/types-of-malware/

Virus: behaviour, evidence and response



1 CORE BEHAVIOUR

Attaches to a host file and executes when the file runs

2 WATCH FOR

Unexpected file changes; antivirus detections

3 BUSINESS EFFECT + FIRST PRIORITY

Corrupts files and uses human execution to continue spreading. Block execution, isolate, restore clean copies.

Classify from multiple signals; isolate safely and preserve evidence before making destructive changes.

Source: https://www.crowdstrike.com/en-us/cybersecurity-101/malware/types-of-malware/

Worm: behaviour, evidence and response

1 CORE BEHAVIOUR

Self-propagates across reachable systems

2 WATCH FOR

Rapid scanning and repeated connections

3 BUSINESS EFFECT + FIRST PRIORITY

Consumes network capacity and compromises many reachable systems quickly. Segment, patch, isolate affected segments.

Classify from multiple signals; isolate safely and preserve evidence before making destructive changes.



Source: https://www.crowdstrike.com/en-us/cybersecurity-101/malware/types-of-malware/

Trojan: behaviour, evidence and response



1 CORE BEHAVIOUR

Disguises malicious capability as legitimate software

2 WATCH FOR

Untrusted installer, new process or persistence

3 BUSINESS EFFECT + FIRST PRIORITY

Creates trusted-looking access for theft, persistence or a second payload. Application control, trusted sources, EDR.

Classify from multiple signals; isolate safely and preserve evidence before making destructive changes.

Source: https://www.crowdstrike.com/en-us/cybersecurity-101/malware/types-of-malware/

Ransomware: behaviour, evidence and response

1 CORE BEHAVIOUR

Encrypts or disrupts systems and demands payment

2 WATCH FOR

File extensions change, ransom note, unavailable services

3 BUSINESS EFFECT + FIRST PRIORITY

Stops operations, threatens disclosure and forces time-critical recovery decisions. Isolate, preserve evidence, activate response and recovery.

Classify from multiple signals; isolate safely and preserve evidence before making destructive changes.



Source: https://www.crowdstrike.com/en-us/cybersecurity-101/malware/types-of-malware/

Spyware: behaviour, evidence and response



1 CORE BEHAVIOUR

Collects activity, credentials or data covertly

2 WATCH FOR

Browser changes, unusual outbound traffic

3 BUSINESS EFFECT + FIRST PRIORITY

Steals activity, credentials and sensitive information while remaining quiet. Remove persistence, reset credentials safely.

Classify from multiple signals; isolate safely and preserve evidence before making destructive changes.

Source: https://www.crowdstrike.com/en-us/cybersecurity-101/malware/types-of-malware/

Rootkit: behaviour, evidence and response

1 CORE BEHAVIOUR

Hides privileged malicious activity

2 WATCH FOR

Integrity anomalies, unexplained privileged changes

3 BUSINESS EFFECT + FIRST PRIORITY

Undermines trust in the operating system and can hide privileged activity. Offline investigation, rebuild trusted baseline.

Classify from multiple signals; isolate safely and preserve evidence before making destructive changes.



Source: https://www.crowdstrike.com/en-us/cybersecurity-101/malware/types-of-malware/

Bot: behaviour, evidence and response



1 CORE BEHAVIOUR

Places a device under remote command

2 WATCH FOR

Beaconing, DDoS or spam activity

3 BUSINESS EFFECT + FIRST PRIORITY

Turns the device into remotely controlled infrastructure for spam, fraud or DDoS. Block command-and-control, isolate, remediate.

Classify from multiple signals; isolate safely and preserve evidence before making destructive changes.

Source: https://www.crowdstrike.com/en-us/cybersecurity-101/malware/types-of-malware/

Logic bomb: behaviour, evidence and response

1 CORE BEHAVIOUR

Triggers a malicious action when a condition is met

2 WATCH FOR

Scheduled or conditional destructive code

3 BUSINESS EFFECT + FIRST PRIORITY

Creates delayed or conditional sabotage that may resemble an ordinary failure. Code review, least privilege, change control.

Classify from multiple signals; isolate safely and preserve evidence before making destructive changes.



Source: https://www.crowdstrike.com/en-us/cybersecurity-101/malware/types-of-malware/

Adware: behaviour, evidence and response



1 CORE BEHAVIOUR

Displays intrusive advertising and may track browsing or alter settings

2 WATCH FOR

Persistent pop-ups, redirects and unwanted extensions

3 BUSINESS EFFECT + FIRST PRIORITY

Degrades user trust and may introduce tracking, redirects or further unwanted software. Remove unwanted software, review permissions, use trusted sources.

Classify from multiple signals; isolate safely and preserve evidence before making destructive changes.

Source: https://www.crowdstrike.com/en-us/cybersecurity-101/malware/types-of-malware/

Keylogger: behaviour, evidence and response

1 CORE BEHAVIOUR

Records keystrokes to capture credentials and sensitive input

2 WATCH FOR

Unexplained input hooks, suspicious processes or credential theft

3 BUSINESS EFFECT + FIRST PRIORITY

Captures secrets at entry, potentially bypassing strong password complexity. Endpoint protection, MFA, safe rebuild and credential reset.

Classify from multiple signals; isolate safely and preserve evidence before making destructive changes.



Source: https://www.crowdstrike.com/en-us/cybersecurity-101/malware/types-of-malware/

Fileless malware: behaviour, evidence and response



1 CORE BEHAVIOUR

Abuses memory and legitimate administration tools instead of relying on a normal file

2 WATCH FOR

Unexpected PowerShell, script hosts, encoded commands or memory activity

3 BUSINESS EFFECT + FIRST PRIORITY

Blends into legitimate administration activity and reduces reliance on obvious files. Restrict scripting, monitor behaviour, least privilege, EDR.

Classify from multiple signals; isolate safely and preserve evidence before making destructive changes.

Source: https://www.crowdstrike.com/en-us/cybersecurity-101/malware/types-of-malware/

Cryptojacker: behaviour, evidence and response

1 CORE BEHAVIOUR

Hijacks computing resources to mine cryptocurrency

2 WATCH FOR

High CPU or cloud cost, overheating and unknown mining connections

3 BUSINESS EFFECT + FIRST PRIORITY

Consumes compute capacity and cloud budget while masking the attack as poor performance. Patch entry points, control scripts, monitor resource use and cloud spend.

Classify from multiple signals; isolate safely and preserve evidence before making destructive changes.



Source: https://www.crowdstrike.com/en-us/cybersecurity-101/malware/types-of-malware/

Ransomware is an operational event

Prepare

Asset inventory, supported systems, least privilege, segmented access and tested offline backups.

Detect

Mass file changes, disabled security tools, unusual admin activity, lateral movement and ransom notes.

Respond

Isolate safely, preserve evidence, activate the incident team, assess scope and recover from trusted sources.

Social engineering targets decisions

Authority

‘The CEO has approved this.’

Urgency

‘Pay in the next ten minutes.’

Fear

‘Your account will be closed.’

Scarcity

‘Only three places remain.’

Familiarity

‘I am your supplier / colleague.’

Secrecy

‘Do not tell anyone yet.’

Phishing: see it in everyday life

1 WHAT IT CAN LOOK LIKE

‘Your parcel is waiting. Pay S$1.20 at parcel-status.example.test within 30 minutes.’

2 SPOT THESE CLUES

Generic urgency, mismatched link, unexpected attachment

3 DO THIS INSTEAD

Use a known route to the service; report the message

How it works: Broad deceptive email seeks clicks, credentials, payments or malware execution



Source: https://www.police.gov.sg/-/media/SPF/Media-Room/Statistics/Annual-Scams-and-Cybercrime-Brief-2025/Ann…

Spear phishing: see it in everyday life



1 WHAT IT CAN LOOK LIKE

‘Hi Mei, I saw your Project Atlas post. Open the attached revised budget before today's client call.’

2 SPOT THESE CLUES

Accurate context paired with an unusual request or channel

3 DO THIS INSTEAD

Verify the request with the known person using another channel

How it works: A tailored message uses role, project or relationship details to appear credible

Source: https://www.police.gov.sg/-/media/SPF/Media-Room/Statistics/Annual-Scams-and-Cybercrime-Brief-2025/Ann…

Whaling: see it in everyday life

1 WHAT IT CAN LOOK LIKE

‘The chairman needs the confidential acquisition file now. Send it to this private address and do not copy anyone.’

2 SPOT THESE CLUES

Legal, finance or board urgency; secrecy; authority pressure

3 DO THIS INSTEAD

Executive verification protocol and delegated dual approval

How it works: A high-value executive or decision-maker is specifically targeted



Source: https://www.police.gov.sg/-/media/SPF/Media-Room/Statistics/Annual-Scams-and-Cybercrime-Brief-2025/Ann…

Business email compromise: see it in everyday life



1 WHAT IT CAN LOOK LIKE

‘We changed banks. Pay invoice INV-2048 to the new account today; reply only to this email.’

2 SPOT THESE CLUES

New bank details, lookalike domain, altered reply-to or unusual timing

3 DO THIS INSTEAD

Independent call-back using stored contacts; dual approval

How it works: An attacker impersonates or takes over a trusted mailbox to change a business transaction

Source: https://www.police.gov.sg/-/media/SPF/Media-Room/Statistics/Annual-Scams-and-Cybercrime-Brief-2025/Ann…

Smishing: see it in everyday life

1 WHAT IT CAN LOOK LIKE

‘Bank alert: your account is locked. Tap bank-verify.example.test now.’

2 SPOT THESE CLUES

Delivery, bank or government lure with shortened link and urgency

3 DO THIS INSTEAD

Open the official app or known website instead of the message link

How it works: SMS or messaging platforms deliver fraudulent links, codes or support requests



Source: https://www.police.gov.sg/-/media/SPF/Media-Room/Statistics/Annual-Scams-and-Cybercrime-Brief-2025/Ann…

Vishing: see it in everyday life



1 WHAT IT CAN LOOK LIKE

‘I'm from the bank fraud team. Read me the six-digit code so I can cancel the transfer.’

2 SPOT THESE CLUES

Caller pressure, request for codes, remote access or immediate payment

3 DO THIS INSTEAD

End the call and dial a known official number

How it works: A caller impersonates authority, support, a bank or a colleague to obtain action or secrets

Source: https://www.police.gov.sg/-/media/SPF/Media-Room/Statistics/Annual-Scams-and-Cybercrime-Brief-2025/Ann…

QR phishing / quishing: see it in everyday life

1 WHAT IT CAN LOOK LIKE

A parking notice says ‘Scan to avoid a fine’, but the QR destination is pay-parking.example.test.

2 SPOT THESE CLUES

Unsolicited QR code near payment, login, parking or document access

3 DO THIS INSTEAD

Preview the destination; use a known app or typed address

How it works: A QR code hides the destination and moves the victim onto a mobile browser



Source: https://www.police.gov.sg/-/media/SPF/Media-Room/Statistics/Annual-Scams-and-Cybercrime-Brief-2025/Ann…

Pretexting: see it in everyday life



1 WHAT IT CAN LOOK LIKE

A caller claims to be a new auditor and asks for the staff list and VPN guide ‘for tomorrow's inspection’.

2 SPOT THESE CLUES

Detailed story but weak entitlement, unusual data request or exception

3 DO THIS INSTEAD

Verify identity, authority and need-to-know separately

How it works: A fabricated role and story creates a reason to disclose data or bypass process

Source: https://www.police.gov.sg/-/media/SPF/Media-Room/Statistics/Annual-Scams-and-Cybercrime-Brief-2025/Ann…

Baiting: see it in everyday life

1 WHAT IT CAN LOOK LIKE

A USB labelled ‘2026 Salary Review’ is left beside the office printer.

2 SPOT THESE CLUES

Free software, found USB media or exclusive content

3 DO THIS INSTEAD

Do not connect unknown media; use approved software sources

How it works: An attractive download, prize or abandoned device tempts the victim to interact



Source: https://www.police.gov.sg/-/media/SPF/Media-Room/Statistics/Annual-Scams-and-Cybercrime-Brief-2025/Ann…

Quid pro quo: see it in everyday life



1 WHAT IT CAN LOOK LIKE

A stranger offers free IT support but asks you to install a remote-control app and share your login code.

2 SPOT THESE CLUES

Unexpected help offer that asks for a password, code or installation

3 DO THIS INSTEAD

Use the official support channel and never disclose authenticators

How it works: A promised benefit or service is exchanged for credentials, access or information

Source: https://www.police.gov.sg/-/media/SPF/Media-Room/Statistics/Annual-Scams-and-Cybercrime-Brief-2025/Ann…

Tech-support scam: see it in everyday life

1 WHAT IT CAN LOOK LIKE

A browser pop-up says ‘Virus detected—call support now’ and the caller asks for remote access.

2 SPOT THESE CLUES

Unsolicited warning, remote-access request and pressure to keep the call secret

3 DO THIS INSTEAD

Disconnect, remove remote access safely and contact genuine support

How it works: Fake support persuades the victim to install remote-control software or pay for a nonexistent problem



Source: https://www.police.gov.sg/-/media/SPF/Media-Room/Statistics/Annual-Scams-and-Cybercrime-Brief-2025/Ann…

Tailgating: see it in everyday life



1 WHAT IT CAN LOOK LIKE

Someone carrying boxes asks you to hold the secure door because ‘my access card is in the car’.

2 SPOT THESE CLUES

Visitor avoids reception or relies on politeness and a carried object

3 DO THIS INSTEAD

Challenge politely or escort the person to reception

How it works: An unauthorised person follows an authorised user through a controlled entrance

Source: https://www.police.gov.sg/-/media/SPF/Media-Room/Statistics/Annual-Scams-and-Cybercrime-Brief-2025/Ann…

Shoulder surfing: see it in everyday life

1 WHAT IT CAN LOOK LIKE

A stranger watches you enter a phone PIN on a crowded train, then follows where you keep the device.

2 SPOT THESE CLUES

Crowded space, exposed screen or sensitive conversation

3 DO THIS INSTEAD

Use privacy awareness, screen positioning and clean-desk habits

How it works: Information is observed directly from screens, keyboards, documents or conversations



Source: https://www.police.gov.sg/-/media/SPF/Media-Room/Statistics/Annual-Scams-and-Cybercrime-Brief-2025/Ann…

Dumpster diving: see it in everyday life



1 WHAT IT CAN LOOK LIKE

Printed customer labels and an old USB drive are placed in ordinary rubbish.

2 SPOT THESE CLUES

Sensitive printouts, labels or storage media in normal waste

3 DO THIS INSTEAD

Secure disposal, shredding and verified media destruction

How it works: Discarded documents or devices reveal identities, systems or business information

Source: https://www.police.gov.sg/-/media/SPF/Media-Room/Statistics/Annual-Scams-and-Cybercrime-Brief-2025/Ann…

Watering-hole deception: see it in everyday life

1 WHAT IT CAN LOOK LIKE

A familiar industry forum suddenly asks you to install a ‘document viewer’ before reading a post.

2 SPOT THESE CLUES

Unexpected login, download or browser warning on a familiar niche site

3 DO THIS INSTEAD

Patch browsers, limit execution and verify unusual site behaviour

How it works: A site trusted by a target community is compromised or imitated to reach that group



Source: https://www.police.gov.sg/-/media/SPF/Media-Room/Statistics/Annual-Scams-and-Cybercrime-Brief-2025/Ann…

MFA fatigue / push bombing: see it in everyday life



1 WHAT IT CAN LOOK LIKE

Your phone shows six login approvals, then ‘IT support’ calls and tells you to approve the next one.

2 SPOT THESE CLUES

Multiple unexpected prompts followed by a call claiming to be support

3 DO THIS INSTEAD

Deny, report, change credentials and prefer number matching or passkeys

How it works: Repeated authentication prompts pressure a user to approve one to make them stop

Source: https://www.police.gov.sg/-/media/SPF/Media-Room/Statistics/Annual-Scams-and-Cybercrime-Brief-2025/Ann…

Phishing email example: parcel delivery

SYNTHETIC CLASSROOM EXAMPLE

EMAIL

MAIL • INBOX • MESSAGE

From **Parcel Updates <notify@parcel-status.example.test>**

Reply-to **claims@parcel-help.example.test**

Subject **Final notice: S$1.20 delivery fee**

Hello,

We could not deliver your parcel. Pay S$1.20 within 30 minutes or it will be returned.

PAY DELIVERY FEE

Mock message — do not click

WHAT SHOULD MAKE YOU PAUSE?

1 **CLUE 1**

Unexpected small payment

2 **CLUE 2**

Sender and reply-to differ

3 **CLUE 3**

Urgency creates pressure

4 **CLUE 4**

Link is not the courier's known site

SAFER RESPONSE

Open the courier's official app or type its known address. Do not use the email link.

Phishing email: genuine or suspicious?

SYNTHETIC CLASSROOM EXAMPLE

EMAIL

MAIL • INBOX • MESSAGE

From **HR Shared Services <benefits@company.example.test>**

Reply-to **benefits-help@external-mail.example.test**

Subject **Urgent: benefits account expires today**

Hello,

Sign in now to keep your medical benefits. You have 20 minutes before access is removed.

KEEP MY BENEFITS

Mock message — do not click

WHAT SHOULD MAKE YOU PAUSE?

1 **CLUE 1**

Threat of losing access

2 **CLUE 2**

Unusual external reply-to

3 **CLUE 3**

Unexpected sign-in link

4 **CLUE 4**

No matching notice in the HR portal

SAFER RESPONSE

Use the bookmarked HR portal or call HR on the directory number.

BEC email example: changed bank details

SYNTHETIC CLASSROOM EXAMPLE

EMAIL

MAIL • INBOX • MESSAGE

From **Lee Supplies <accounts@lee-supp1ies.example.test>**

Reply-to **payments@lee-supp1ies.example.test**

Subject **RE: INV-2048 — new bank account**

Hello,

Our bank account changed today. Please pay S$48,700 to the attached account before 3 pm. Reply only to this email.

OPEN NEW BANK DETAILS

Mock message — do not click

WHAT SHOULD MAKE YOU PAUSE?

1 **CLUE 1**

Lookalike domain uses the digit 1

2 **CLUE 2**

Bank details changed

3 **CLUE 3**

Same-day payment pressure

4 **CLUE 4**

Request bypasses normal verification

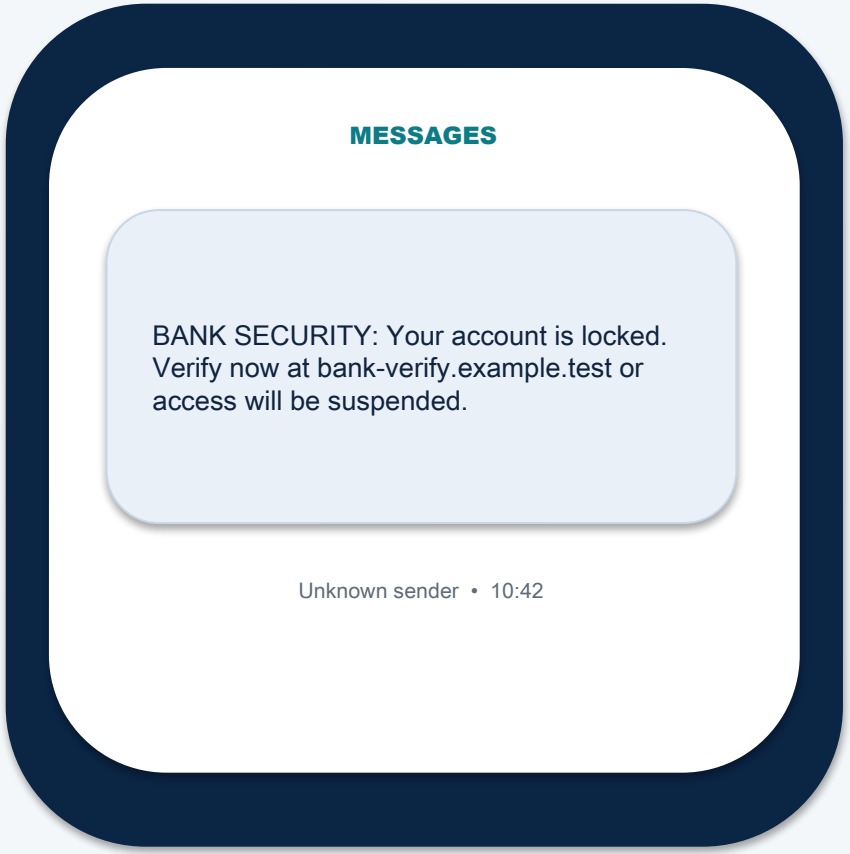
SAFER RESPONSE

Stop payment and call the supplier using the number already stored in the vendor record.

Smishing example: account locked

SYNTHETIC CLASSROOM EXAMPLE

SMS



WHAT SHOULD MAKE YOU PAUSE?

1 **CLUE 1**

Fear of account loss

2 **CLUE 2**

Link asks for a bank login

3 **CLUE 3**

Generic sender name

4 **CLUE 4**

Message is unexpected

SAFER RESPONSE

Open the official banking app independently and use its support number if needed.

Vishing example: the fake fraud-team call

SYNTHETIC CLASSROOM EXAMPLE

PHONE CALL

?

INCOMING CALL

‘Bank Fraud Team’

Caller ID can be spoofed

Caller: ‘There is a S$2,900 transfer pending. Read me the six-digit code now so I can cancel it.’

The bank will not need your one-time code to cancel a transfer.

WHAT SHOULD MAKE YOU PAUSE?

1 **CLUE 1**

Caller creates fear

2 **CLUE 2**

Requests an authentication code

3 **CLUE 3**

Demands immediate action

4 **CLUE 4**

Caller ID can be spoofed

SAFER RESPONSE

End the call. Dial the number on the back of the bank card or in the official app.

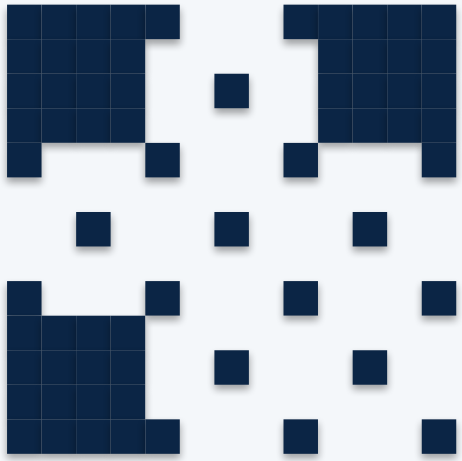
QR phishing example: scan to pay

SYNTHETIC CLASSROOM EXAMPLE

QR POSTER

PARKING NOTICE

SCAN NOW TO AVOID A S$70 FINE



NON-SCANNABLE MOCK QR

Preview: pay-parking.example.test

WHAT SHOULD MAKE YOU PAUSE?

1 **CLUE 1**

Sticker may cover a genuine code

2 **CLUE 2**

Destination is hidden until scanning

3 **CLUE 3**

Fine creates urgency

4 **CLUE 4**

Site is not the known parking operator

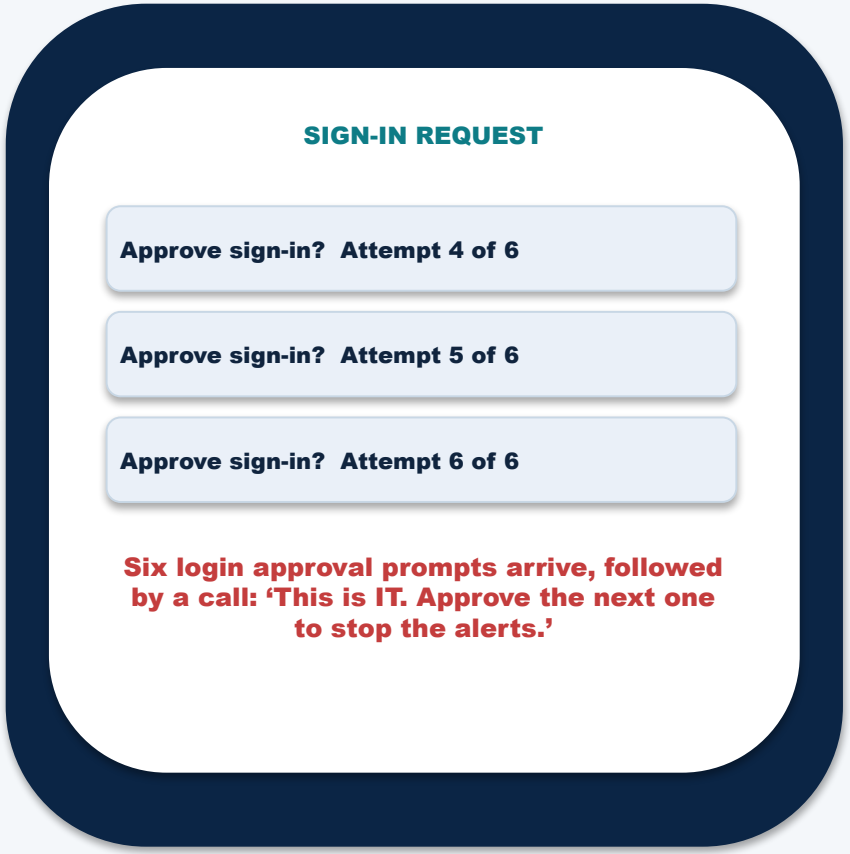
SAFER RESPONSE

Use the official parking app or type the operator's known address.

MFA fatigue example: repeated prompts

SYNTHETIC CLASSROOM EXAMPLE

AUTHENTICATION APP



WHAT SHOULD MAKE YOU PAUSE?

1 **CLUE 1**

You did not start a login

2 **CLUE 2**

Prompts repeat

3 **CLUE 3**

A caller asks you to approve

4 **CLUE 4**

Approval would authorise the attacker

SAFER RESPONSE

Deny, report the prompts, change the password from a trusted device and review active sessions.

Physical social engineering examples

SYNTHETIC CLASSROOM EXAMPLE

WORKPLACE

1 TAILGATING

A person carrying boxes asks you to hold a secure door: ‘My access card is in the car.’

2 PRETEXTING

A ‘new auditor’ asks for the staff list and VPN guide before an unverified inspection.

A convincing story is not proof of identity or authority.

WHAT SHOULD MAKE YOU PAUSE?

1 CLUE 1

Politeness is used as pressure

2 CLUE 2

No visible access or appointment

3 CLUE 3

Role sounds plausible but is unverified

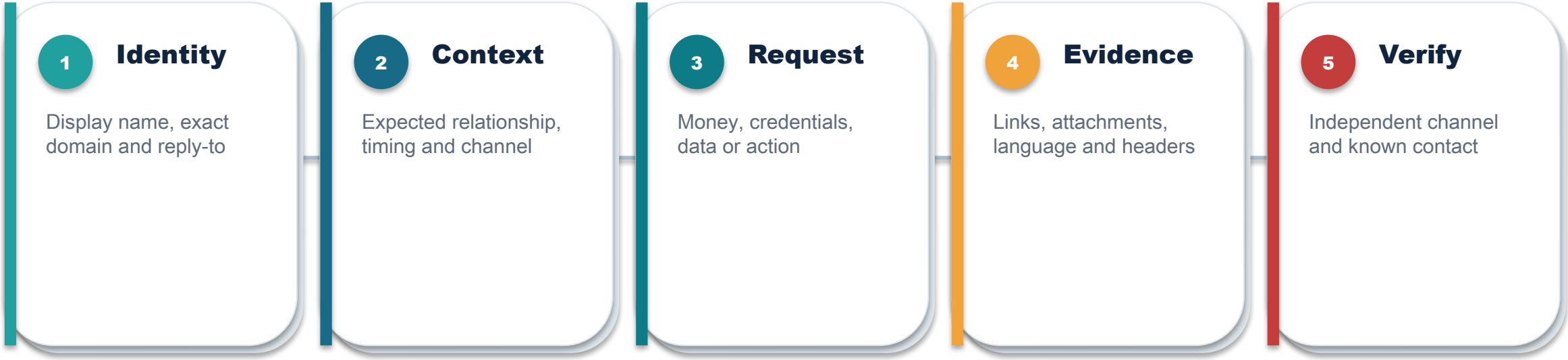
4 CLUE 4

Request exceeds need-to-know

SAFER RESPONSE

Escort the person to reception and verify identity, appointment, authority and need-to-know.

Phishing and BEC: inspect the whole request



Do not use contact details supplied inside the suspicious message.

SAME PSYCHOLOGY • DIFFERENT SURFACE

Scams move across channels

Smishing

SMS or messaging link designed to steal data or install software.

Vishing

Voice call creates urgency or impersonates authority.

QR phishing

A code hides the destination and moves the user to mobile.

Social account takeover

A familiar identity asks for money, codes or access.

E-commerce fraud

Fake listing, payment page, refund or delivery notice.

Tech-support scam

Attacker persuades the user to install remote-control software.

Networks face passive and active attacks

Passive observation

Traffic analysis or eavesdropping seeks information without changing the system.

Man-in-the-middle

An attacker intercepts or alters communication between trusted endpoints.

Spoofing

A source identity, address or service is imitated to gain trust.

DDoS

Many sources exhaust service capacity and reduce availability.

Rogue wireless

A fake or unsafe access point captures traffic or credentials.

Lateral movement

Compromise spreads through excessive trust between systems.

DDoS is an availability attack

WHAT HAPPENS

- Traffic or requests overwhelm capacity
- Legitimate users cannot reach the service
- Attack may distract from another intrusion
- Cloud and upstream support may be required

RESILIENCE

- Rate limits and filtering
- Content delivery and traffic scrubbing
- Capacity and failover planning
- Provider escalation runbook

Source: https://www.cisco.com/site/us/en/learn/topics/security/what-is-cybersecurity.html

Mobile devices concentrate identity and access

Risks

Malicious apps, unsafe permissions, lost device, SIM swap, QR phishing and untrusted Wi-Fi.

Baseline

Supported OS, automatic updates, screen lock, encryption, remote locate/wipe and safe app sources.

Judgement

Treat unexpected MFA prompts and recovery messages as potential account takeover.

Nation-state and critical-infrastructure risk

Patient access

Long dwell time and low-noise collection can matter more than immediate disruption.

Trusted pathways

Suppliers, identities and management tools offer broad reach.

Civil consequence

Essential services connect cyber resilience with public safety and national security.

Individual role

Reporting anomalies and protecting credentials still breaks high-end attack chains.

Security, privacy and lawful access

PROTECTION

- Encryption protects confidentiality in transit and at rest
- Access controls limit who can use data
- Logging creates accountability
- Data minimisation reduces exposure

GOVERNANCE

- Use must be authorised and proportionate
- Retention and monitoring need policy
- Escalate legal or ethical uncertainty
- Do not bypass controls for convenience

SingHealth data breach

Singapore, 2018

1 ENTRY

Compromised front-end workstation and privileged credentials

2 WEAKNESS

Endpoint compromise, credential misuse, delayed detection

3 IMPACT

Personal particulars of 1.5 million patients; outpatient medication records of 160,000

WHAT DEFENDERS COULD SEE

Unusual database queries, privileged access, suspicious workstation activity

CONTROLS THAT BREAK THE CHAIN

Endpoint hardening, least privilege, privileged access monitoring, segmented systems, rehearsed response

TRANSFERABLE LESSON

Identity and endpoint signals must be connected before an intruder reaches sensitive data.

SOURCE: Ministry of Health Singapore • SingHealth's IT System Target of Cyberattack • moh.gov.sg • CLICK TO OPEN OFFICIAL SOURCE

Source: https://www.moh.gov.sg/newsroom/singhealth%27s-it-system-target-of-cyberattack/

Decision room: SingHealth data breach

TEAM DISCUSSION • 8 MIN

1

TRIAGE

Which evidence would you trust first: Unusual database queries, privileged access, suspicious workstation activity?

2

DECISION

Which immediate action limits personal particulars of 1.5 million patients; outpatient medication records of 160,000 without destroying evidence?

3

CONTROL

Which one control should be verified first from: Endpoint hardening, least privilege, privileged access monitoring, segmented systems, rehearsed response?

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TRANSFER

Where could the same entry path exist in a personal account or small business today?



SOURCE: Ministry of Health Singapore • SingHealth's IT System Target of Cyberattack • moh.gov.sg • CLICK TO OPEN OFFICIAL SOURCE

Source: https://www.moh.gov.sg/newsroom/singhealth%27s-it-system-target-of-cyberattack/

WannaCry and the NHS

United Kingdom, 2017

1 ENTRY

Ransomware exploited unpatched Windows systems

2 WEAKNESS

Patch gaps, unsupported systems, uneven asset visibility

3 IMPACT

Disruption across NHS organisations and cancelled care

WHAT DEFENDERS COULD SEE

Rapid lateral spread, inaccessible files, ransom notes, unavailable clinical systems

CONTROLS THAT BREAK THE CHAIN

Asset inventory, timely patching, supported platforms, segmentation, tested offline recovery

TRANSFERABLE LESSON

A known vulnerability becomes an operational crisis when ownership and patching fail.

SOURCE: UK National Audit Office • Investigation: WannaCry Cyber Attack and the NHS • nao.org.uk • CLICK TO OPEN OFFICIAL SOURCE

Source: https://www.nao.org.uk/reports/investigation-wannacry-cyber-attack-and-the-nhs/

Decision room: WannaCry and the NHS

TEAM DISCUSSION • 8 MIN

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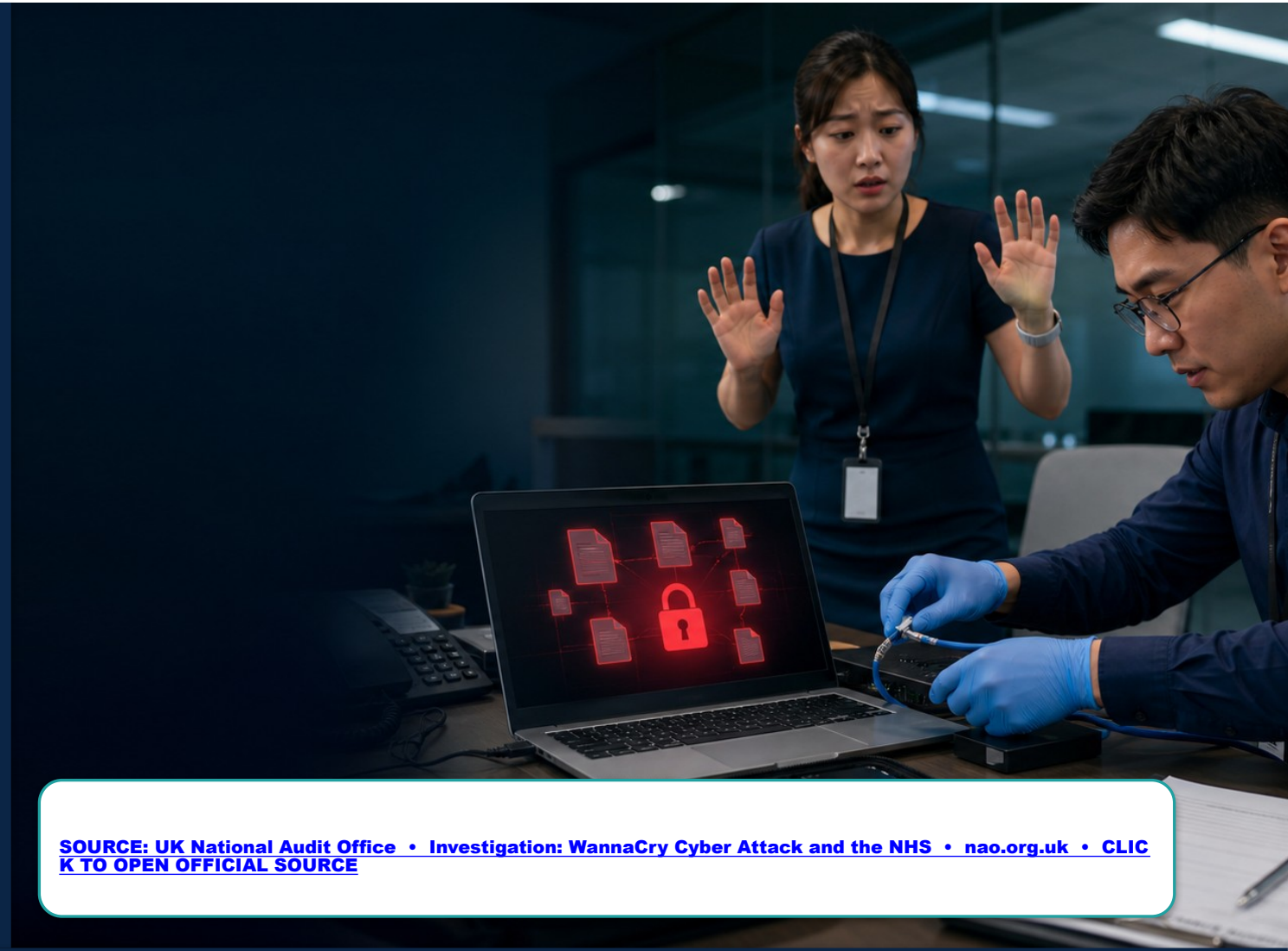
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SOURCE: UK National Audit Office • Investigation: WannaCry Cyber Attack and the NHS • nao.org.uk • CLICK TO OPEN OFFICIAL SOURCE

Source: https://www.nao.org.uk/reports/investigation-wannacry-cyber-attack-and-the-nhs/

Equifax breach

United States, 2017

1 ENTRY

Exploitation of a known web application vulnerability

2 WEAKNESS

Patch and asset-management breakdowns

3 IMPACT

Sensitive personal information exposed at massive scale

WHAT DEFENDERS COULD SEE

Unusual web traffic and data access; vulnerable internet-facing service

CONTROLS THAT BREAK THE CHAIN

Complete asset inventory, vulnerability SLAs, compensating controls, verification after patching

TRANSFERABLE LESSON

A patch notice is not a control until the affected asset is found, fixed and verified.

SOURCE: U.S. House Committee on Oversight • The Equifax Data Breach • oversight.house.gov • CLICK TO OPEN OFFICIAL SOURCE

Source: https://oversight.house.gov/report/the-equifax-data-breach/

Decision room: Equifax breach

TEAM DISCUSSION • 8 MIN

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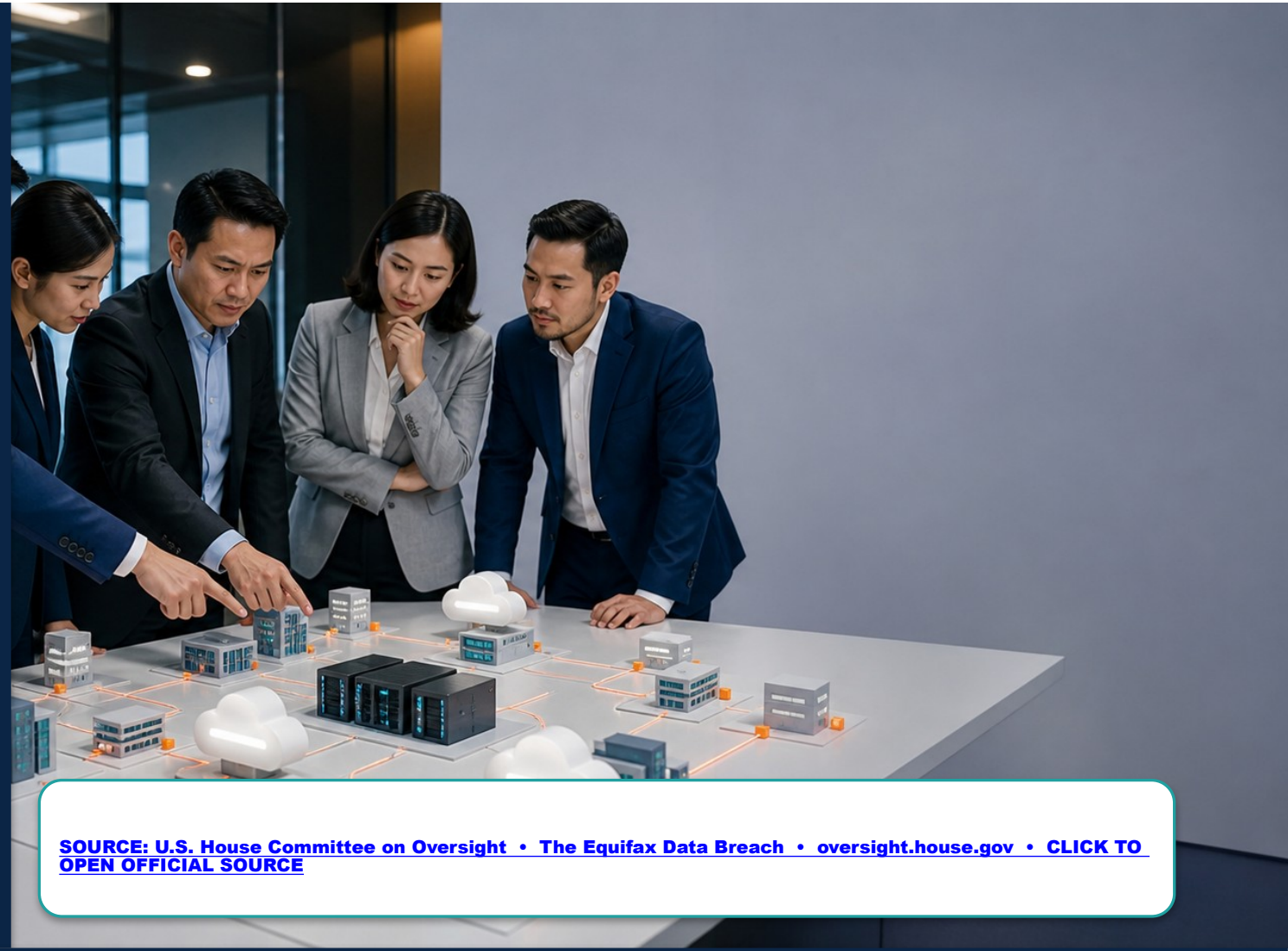
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SolarWinds supply-chain compromise

Global, 2020

1 ENTRY

Malicious code inserted into trusted software updates

2 WEAKNESS

Implicit trust in supplier software and privileged management tooling

3 IMPACT

Covert access to public- and private-sector networks

WHAT DEFENDERS COULD SEE

Rare identity use, unusual cloud access, unexpected network paths

CONTROLS THAT BREAK THE CHAIN

Supplier risk management, zero-trust checks, identity analytics, egress monitoring, rapid key rotation

TRANSFERABLE LESSON

Trusted software can carry risk; verify behaviour after deployment, not only the publisher.

SOURCE: CISA • Active Exploitation of SolarWinds Software • cisa.gov • CLICK TO OPEN OFFICIAL SOURCE

Source: https://www.cisa.gov/news-events/alerts/2020/12/13/active-exploitation-solarwinds-software

Decision room: SolarWinds supply-chain compromise

TEAM DISCUSSION • 8 MIN

1

TRIAGE

Which evidence would you trust first: Rare identity use, unusual cloud access, unexpected network paths?

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DECISION

Which immediate action limits covert access to public- and private-sector networks without destroying evidence?

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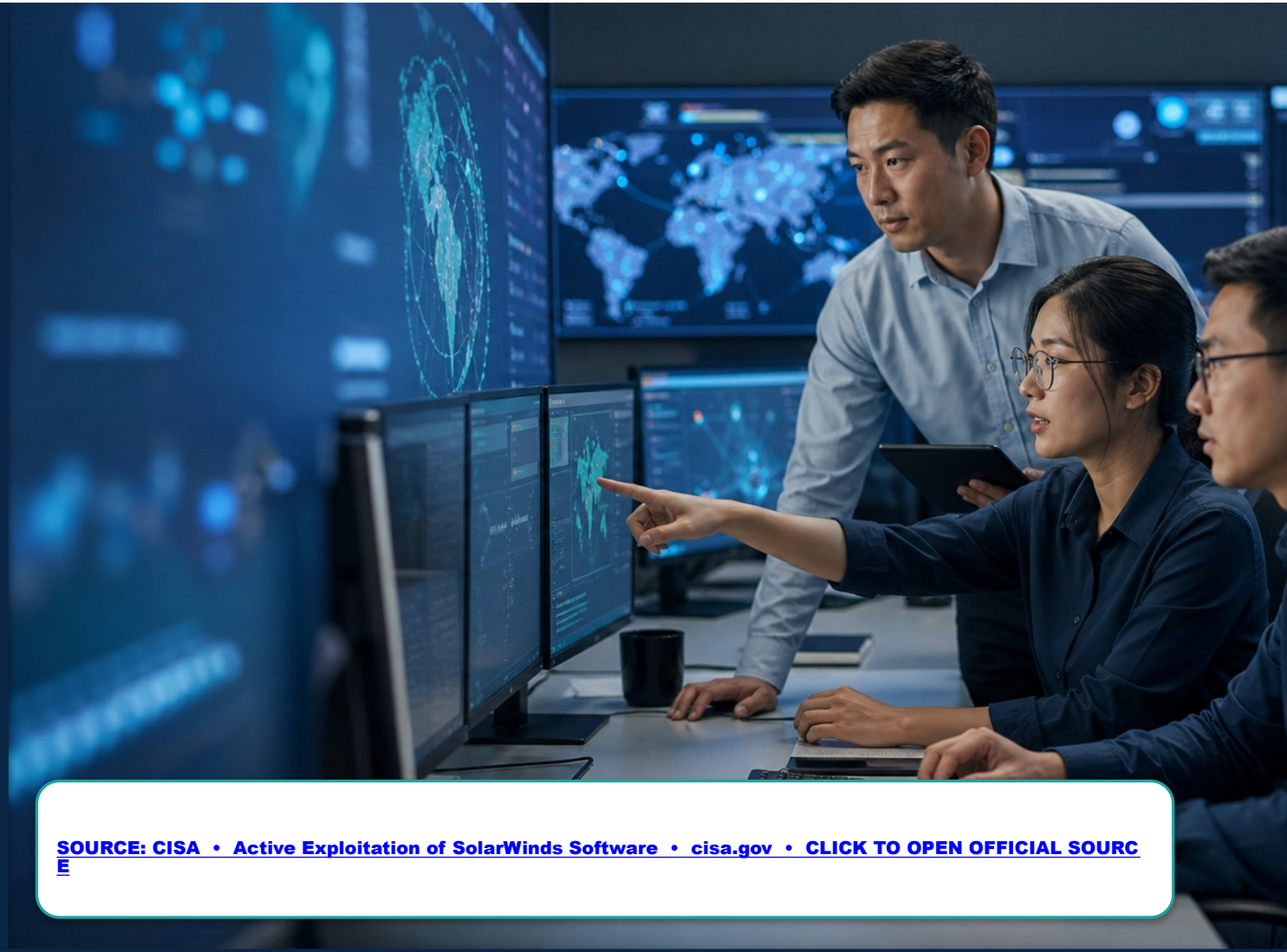
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Colonial Pipeline

United States, 2021

1 ENTRY

Compromised VPN password on an inactive account without MFA

2 WEAKNESS

Dormant identity, password reuse/exposure, missing MFA

3 IMPACT

Business systems disruption and precautionary pipeline shutdown

WHAT DEFENDERS COULD SEE

Unexpected remote access and activity from a dormant account

CONTROLS THAT BREAK THE CHAIN

Disable dormant accounts, MFA, password monitoring, remote-access review, IT/OT separation

TRANSFERABLE LESSON

Identity hygiene is infrastructure resilience.

SOURCE: CISA • IAM Case Study: Colonial Pipeline • cisa.gov • CLICK TO OPEN OFFICIAL SOURCE

Decision room: Colonial Pipeline

TEAM DISCUSSION • 8 MIN

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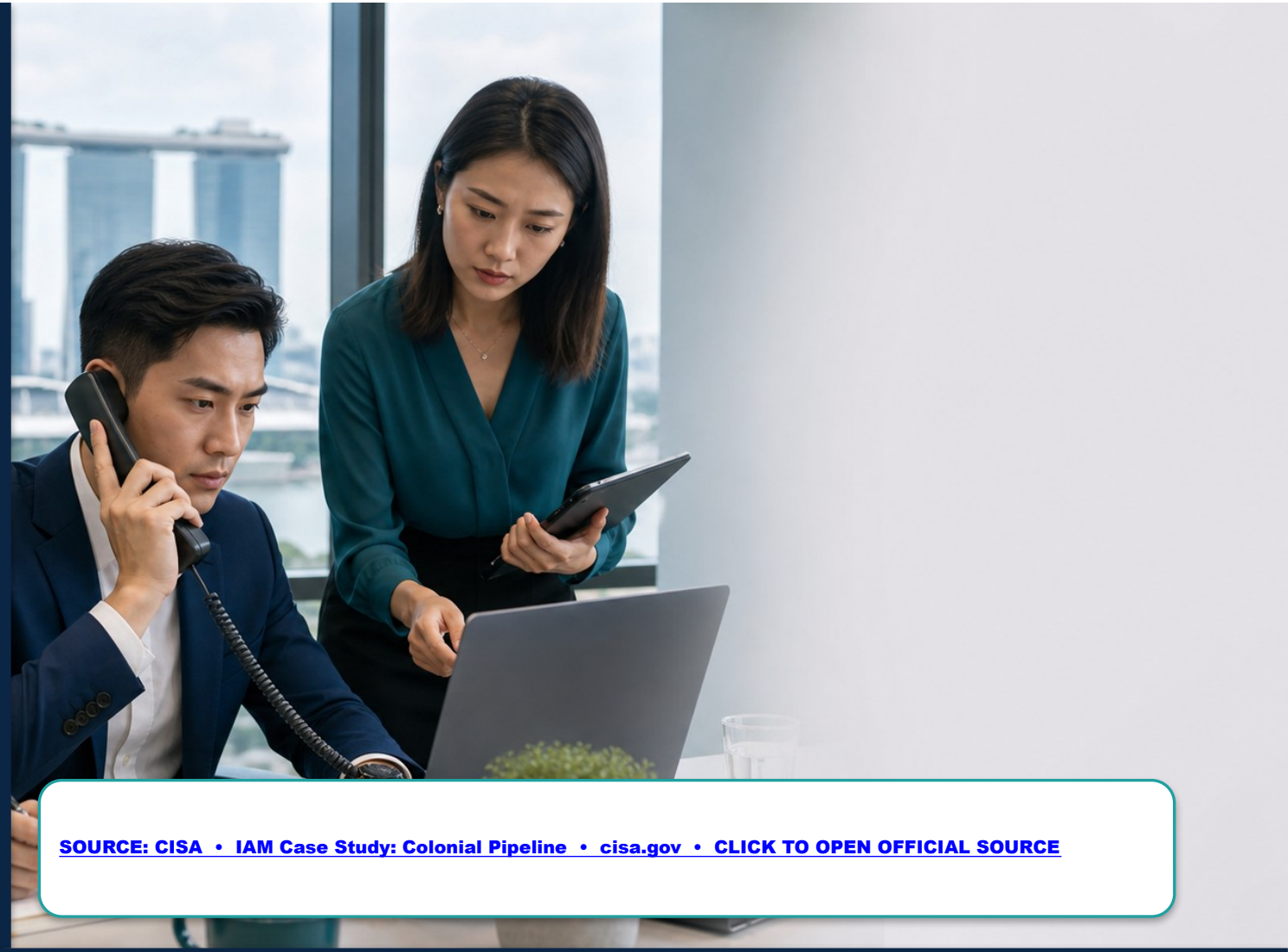
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SOURCE: CISA • IAM Case Study: Colonial Pipeline • cisa.gov • CLICK TO OPEN OFFICIAL SOURCE

Source: https://www.cisa.gov/sites/default/files/2023-12/IAM_EDUCATIONAL_AID_TALKING_POINTS_508C.pdf

Log4Shell exploitation

Global, 2021

1 ENTRY

Remote code execution through a widely embedded logging component

2 WEAKNESS

Hidden software dependency and incomplete inventory

3 IMPACT

Broad exploitation risk across internet-facing systems

WHAT DEFENDERS COULD SEE

Suspicious lookup strings, outbound callbacks, new processes and payloads

CONTROLS THAT BREAK THE CHAIN

Software bill of materials, emergency patch process, virtual patching, threat hunting

TRANSFERABLE LESSON

You cannot protect a component you do not know you run.

SOURCE: CISA • Mitigating Log4Shell and Log4j Vulnerabilities • cisa.gov • CLICK TO OPEN OFFICIAL SOURCE

Decision room: Log4Shell exploitation

TEAM DISCUSSION • 8 MIN

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Source: https://www.cisa.gov/news-events/cybersecurity-advisories/aa21-356a

MOVEit mass exploitation

Global, 2023

1 ENTRY

Zero-day SQL injection in managed file-transfer software

2 WEAKNESS

Internet-facing data service and third-party concentration

3 IMPACT

Data theft across many organisations through one common platform

WHAT DEFENDERS COULD SEE

Unexpected database queries, web shells, unusual downloads and new accounts

CONTROLS THAT BREAK THE CHAIN

Vendor advisories, rapid isolation, compromise assessment, third-party exposure mapping

TRANSFERABLE LESSON

Third-party software can multiply one vulnerability into sector-wide exposure.

SOURCE: CISA • CL0P Exploits MOVEit Vulnerability • cisa.gov • CLICK TO OPEN OFFICIAL SOURCE

Source: https://www.cisa.gov/sites/default/files/2023-06/aa23-158a-stopransomware-cl0p-ransomware-gang-exploit…

Decision room: MOVEit mass exploitation

TEAM DISCUSSION • 8 MIN

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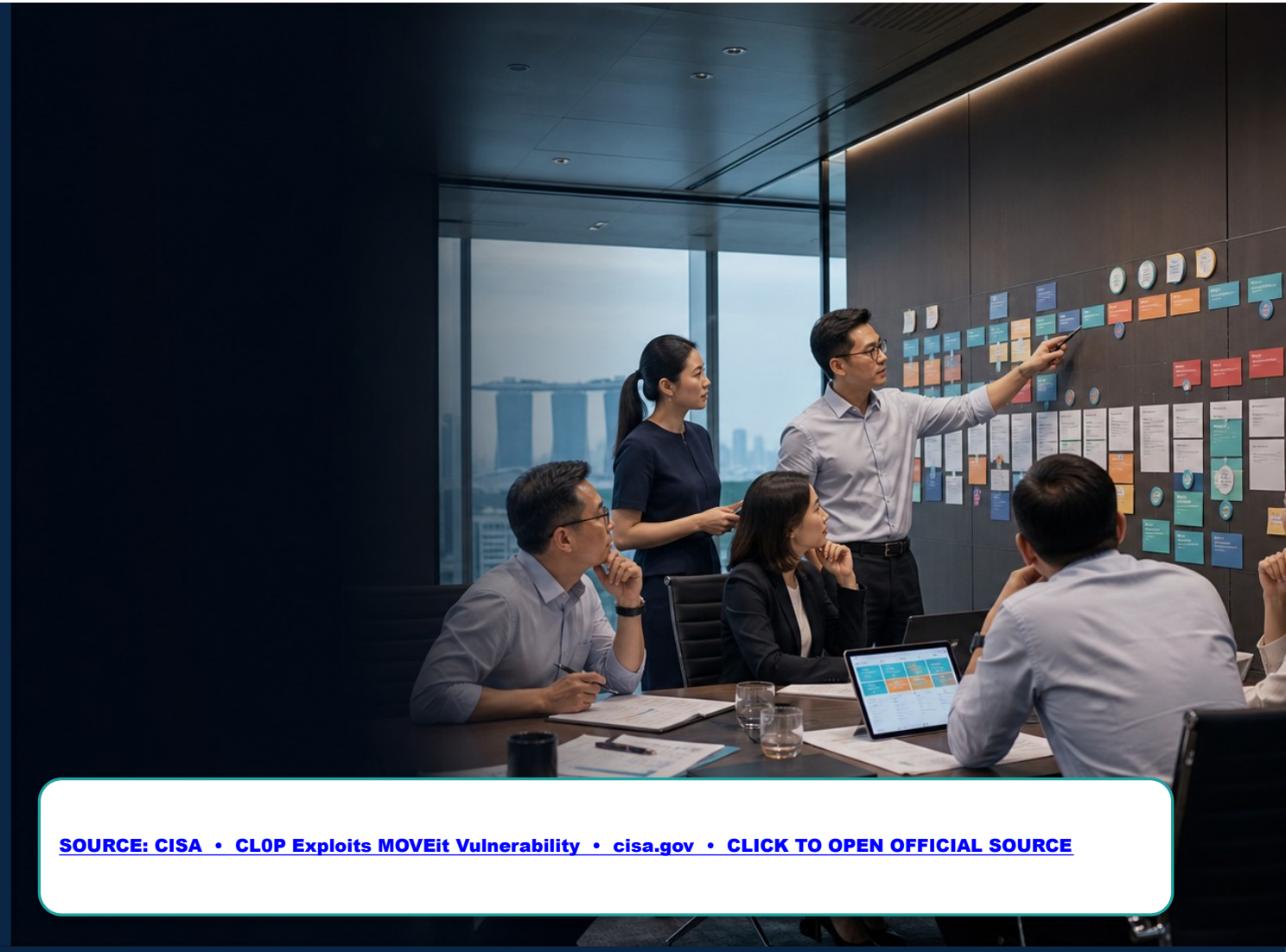
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MGM Resorts cyber incident

United States, 2023

1 ENTRY

Identity compromise followed by system disruption

2 WEAKNESS

Help-desk and identity-recovery processes exposed to manipulation

3 IMPACT

Systems shut down, service disruption and personal data exposure

WHAT DEFENDERS COULD SEE

Unusual identity recovery, privileged account changes and abrupt service outages

CONTROLS THAT BREAK THE CHAIN

High-assurance help-desk verification, phishing-resistant MFA, privileged identity monitoring

TRANSFERABLE LESSON

Recovery workflows need controls as strong as login workflows.

SOURCE: U.S. Securities and Exchange Commission • MGM Resorts International Form 8-K • sec.gov • CLICK TO OPEN OFFICIAL SOURCE

Source: https://www.sec.gov/Archives/edgar/data/789570/000119312523251667/d461062d8k.htm

Decision room: MGM Resorts cyber incident

TEAM DISCUSSION • 8 MIN

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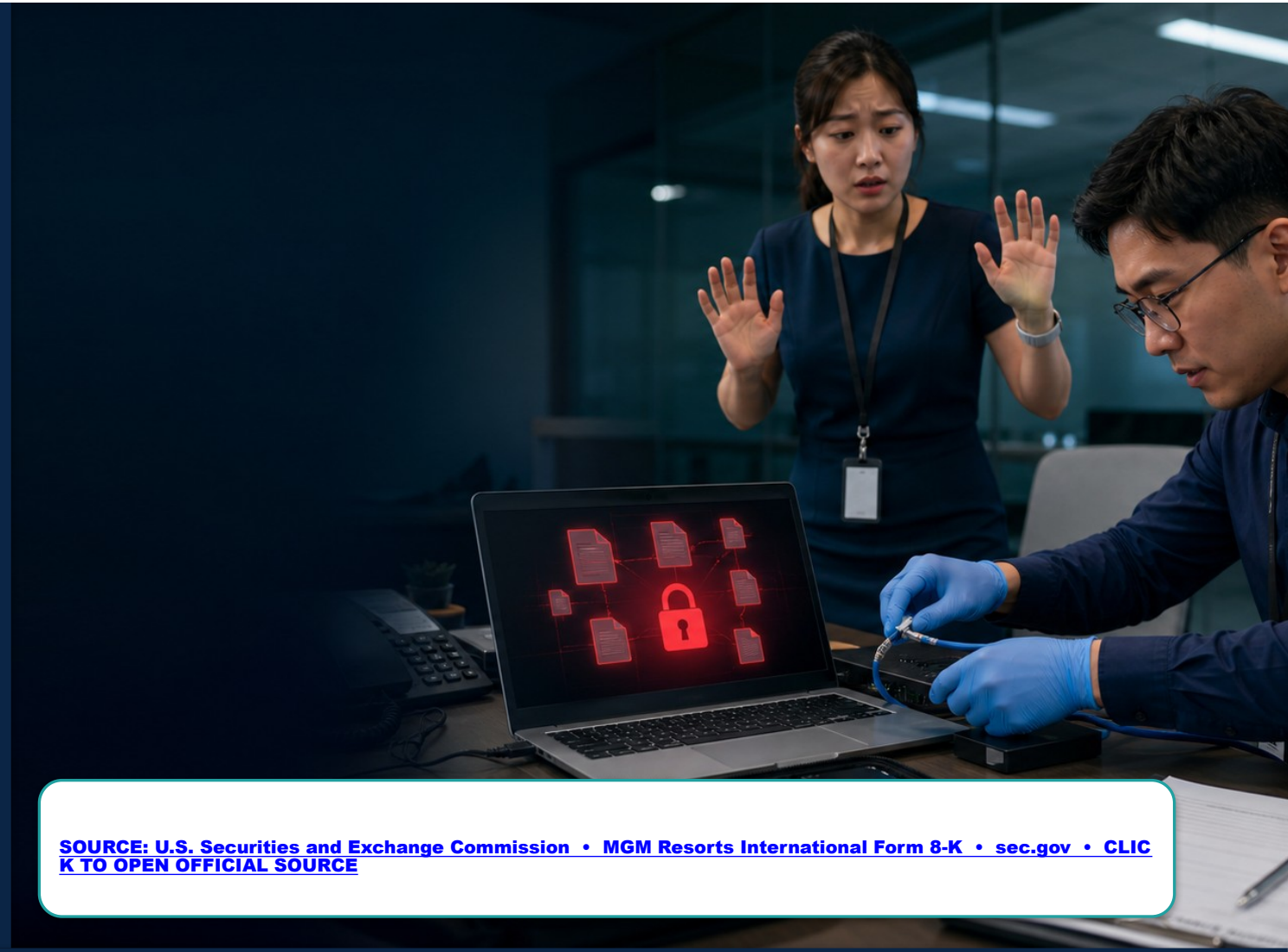
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Singapore commodity-trader BEC

Singapore, 2026

1 ENTRY

Supplier impersonation using a subtly transposed email domain

2 WEAKNESS

Payment change accepted without independent verification

3 IMPACT

USD 6.6 million transferred; funds recovered after rapid reporting and international cooperation

WHAT DEFENDERS COULD SEE

Changed bank details, lookalike domain, urgency and unusual payment instructions

CONTROLS THAT BREAK THE CHAIN

Call-back verification using known contacts, dual approval, domain checks, rapid bank and police escalation

TRANSFERABLE LESSON

A two-minute independent check can prevent a multi-million-dollar loss.

SOURCE: Singapore Police Force • Recovery of Over USD 6.6 Million in BEC Case • police.gov.sg • CLICK TO OPEN OFFICIAL SOURCE

Source: https://www.police.gov.sg/media-hub/news/2026/05/20260505_international_cooperation_leads_to_the_recov…

Decision room: Singapore commodity-trader BEC

TEAM DISCUSSION • 8 MIN

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Source: https://www.police.gov.sg/media-hub/news/2026/05/20260505_international_cooperation_leads_to_the_recov…

Target payment-card breach

United States, 2013

1 ENTRY

Stolen third-party vendor credentials enabled access into the retailer environment

2 WEAKNESS

Supplier access, weak separation and missed opportunities to act on security alerts

3 IMPACT

Payment-card information and personal data associated with tens of millions of customers were exposed

WHAT DEFENDERS COULD SEE

Malware alerts, unusual internal movement and outbound data transfers

CONTROLS THAT BREAK THE CHAIN

Supplier MFA, segmented vendor access, alert ownership, egress monitoring and tested incident escalation

TRANSFERABLE LESSON

A low-privilege supplier pathway can become a high-impact route when trust boundaries are weak.

SOURCE: U.S. Senate Committee on Commerce • Target's Missed Opportunities to Stop a Data Breach • commerce.senate.gov • CLICK TO OPEN OFFICIAL SOURCE

Source: https://www.commerce.senate.gov/press/dem/release/rockefeller-staff-report-details-targets-missed-oppo…

Decision room: Target payment-card breach

TEAM DISCUSSION • 8 MIN

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Source: https://www.commerce.senate.gov/press/dem/release/rockefeller-staff-report-details-targets-missed-oppo…

NotPetya disruption at Maersk

Global shipping, 2017

1 ENTRY

Destructive malware entered through compromised Ukrainian accounting software and spread globally

2 WEAKNESS

Trusted software pathway, flat connectivity and operational dependence on shared digital services

3 IMPACT

Global applications and data were unavailable for days; port and shipping operations were disrupted

WHAT DEFENDERS COULD SEE

Rapid cross-site outages, unavailable systems and destructive behaviour rather than a recoverable ransom event

CONTROLS THAT BREAK THE CHAIN

Supplier risk controls, segmentation, isolated recovery capability, offline identity services and business workarounds

TRANSFERABLE LESSON

An organisation can suffer severe collateral damage even when it is not the intended target.

SOURCE: UK Department for Transport • Cyber Security for Ports and Port Systems • assets.publishing.service.gov.uk • CLICK TO OPEN OFFICIAL SOURCE

Source: https://assets.publishing.service.gov.uk/government/uploads/system/uploads/attachment_data/file/546160…

Decision room: NotPetya disruption at Maersk

TEAM DISCUSSION • 8 MIN

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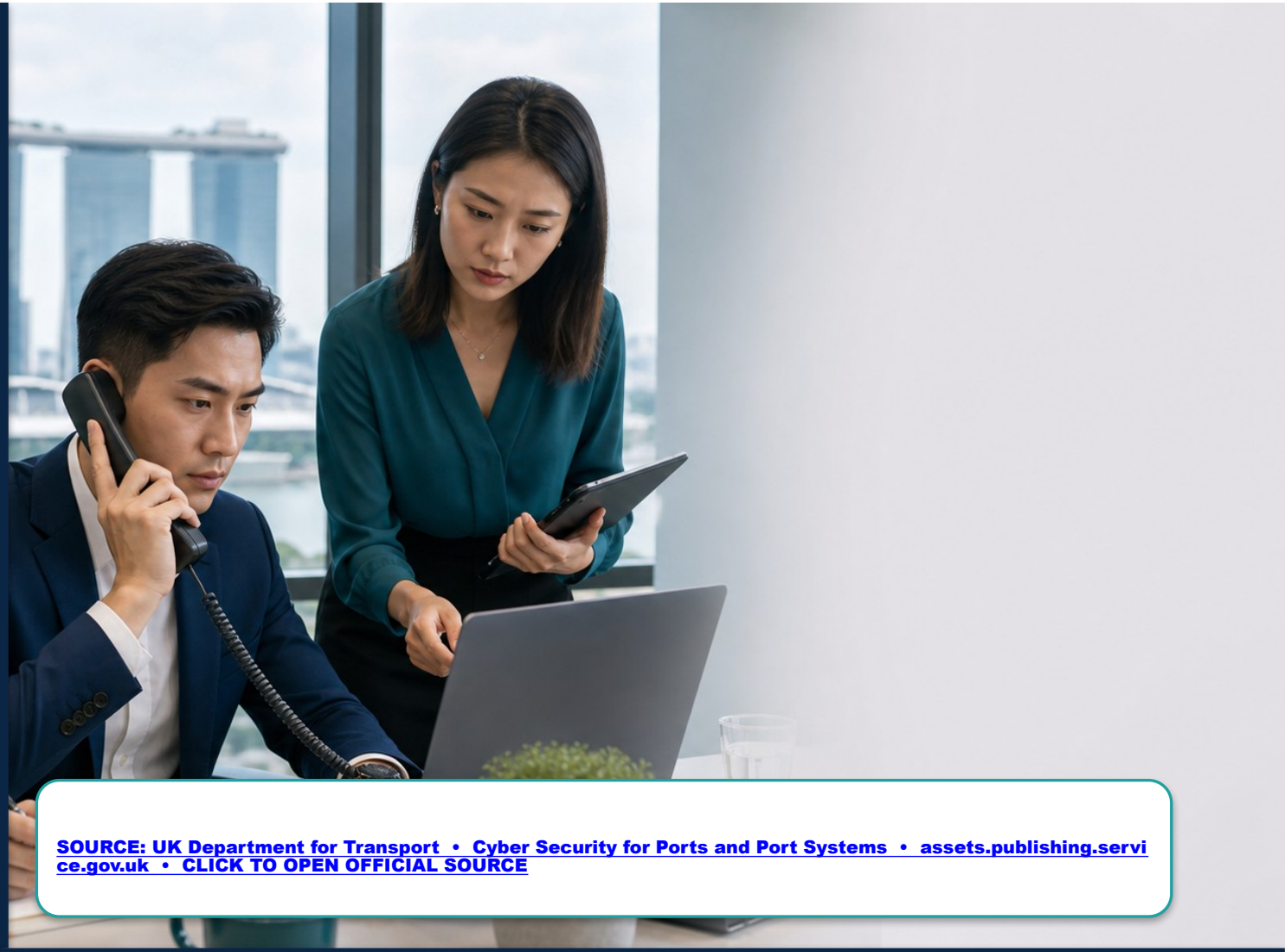
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Source: https://assets.publishing.service.gov.uk/government/uploads/system/uploads/attachment_data/file/546160…

Marriott Starwood data breach

Global hospitality, disclosed 2018

1 ENTRY

Long-running unauthorised access in an acquired reservation environment

2 WEAKNESS

Acquisition due diligence, legacy-system monitoring and delayed detection

3 IMPACT

Large-scale exposure of guest records and sustained regulatory and customer consequences

WHAT DEFENDERS COULD SEE

Compromised accounts, database activity and anomalous access within a legacy environment

CONTROLS THAT BREAK THE CHAIN

Pre-acquisition security assurance, rapid integration, identity reset, continuous monitoring and data minimisation

TRANSFERABLE LESSON

Cyber risk transfers with an acquisition unless inherited systems and identities are independently verified.

SOURCE: UK Information Commissioner's Office • Marriott International Monetary Penalty Notice • ico.org.uk • CLICK TO OPEN OFFICIAL SOURCE

Source: https://ico.org.uk/media/action-weve-taken/mpns/2618524/marriott-international-inc-mpn-20201030.pdf

Decision room: Marriott Starwood data breach

TEAM DISCUSSION • 8 MIN

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Source: https://ico.org.uk/media/action-weve-taken/mpns/2618524/marriott-international-inc-mpn-20201030.pdf

Capital One cloud data breach

United States, 2019

1 ENTRY

A misconfigured web application firewall enabled access to cloud-hosted data

2 WEAKNESS

Cloud configuration and permission design allowed excessive access

3 IMPACT

Data from millions of credit applications was accessed

WHAT DEFENDERS COULD SEE

Unexpected cloud API activity, unusual data access and exploitation of an internet-facing control

CONTROLS THAT BREAK THE CHAIN

Least-privilege cloud roles, configuration assurance, logging, anomaly detection and rapid credential revocation

TRANSFERABLE LESSON

Cloud services inherit the organisation's configuration decisions; provider security does not remove customer responsibility.

SOURCE: U.S. Department of Justice • Seattle Tech Worker Arrested for Data Theft • justice.gov • CLICK TO OPEN OFFICIAL SOURCE

Source: https://www.justice.gov/usao-wdwa/pr/seattle-tech-worker-arrested-data-theft-involving-large-financial…

Decision room: Capital One cloud data breach

TEAM DISCUSSION • 8 MIN

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Source: https://www.justice.gov/usao-wdwa/pr/seattle-tech-worker-arrested-data-theft-involving-large-financial…

Norsk Hydro LockerGoga ransomware

Norway and global operations, 2019

1 ENTRY

Ransomware disrupted corporate and operational workstations

2 WEAKNESS

Enterprise connectivity allowed IT disruption to affect industrial operations

3 IMPACT

Plants shifted to manual workarounds while systems were rebuilt and restored

WHAT DEFENDERS COULD SEE

Unavailable workstations, encryption activity and loss of normal production-support systems

CONTROLS THAT BREAK THE CHAIN

IT/OT segmentation, rehearsed manual operations, clean rebuild capability, transparent communications and tested backups

TRANSFERABLE LESSON

Operational resilience includes safe manual workarounds, not only technical restoration.

SOURCE: UK National Cyber Security Centre • What Is OT Malware? • ncsc.gov.uk • CLICK TO OPEN OFFICIAL SOURCE

Source: https://www.ncsc.gov.uk/blog-post/what-is-ot-malware

Decision room: Norsk Hydro LockerGoga ransomware

TEAM DISCUSSION • 8 MIN

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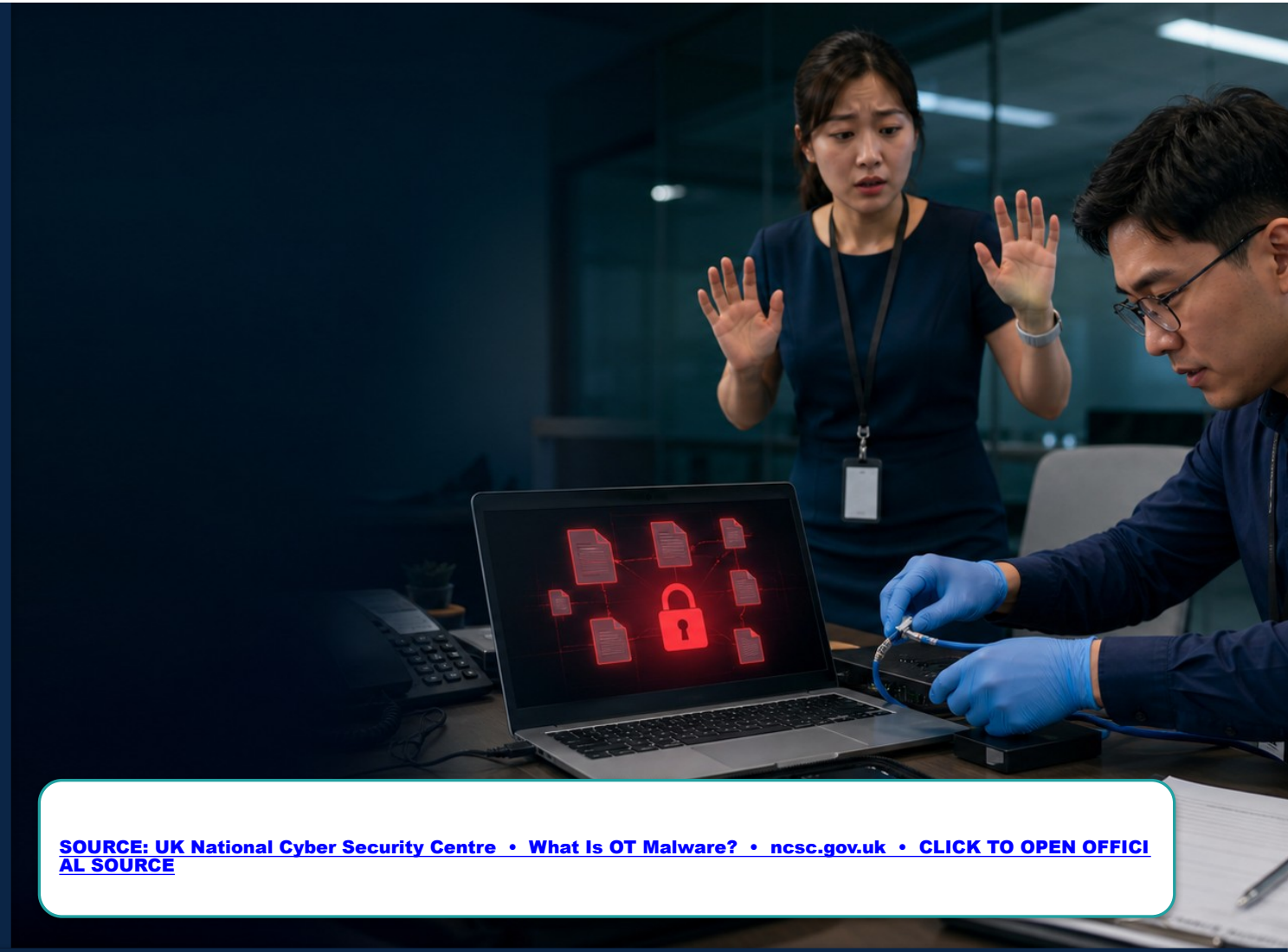
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TRANSFER

Where could the same entry path exist in a personal account or small business today?



SOURCE: UK National Cyber Security Centre • What Is OT Malware? • ncsc.gov.uk • CLICK TO OPEN OFFICIAL SOURCE

Source: https://www.ncsc.gov.uk/blog-post/what-is-ot-malware

Change Healthcare cyberattack

United States healthcare, 2024

1 ENTRY

Threat actor access led the organisation to isolate connected transaction systems

2 WEAKNESS

Sector-wide dependence on a concentrated claims and payment intermediary

3 IMPACT

Electronic claims, payments and pharmacy transactions were disrupted across healthcare providers

WHAT DEFENDERS COULD SEE

Service interruption across connected customers and loss of access to transactional networks

CONTROLS THAT BREAK THE CHAIN

Strong remote-access identity, supplier concentration analysis, alternate clearing routes, isolation plans and recovery communications

TRANSFERABLE LESSON

A cyber incident at one digital intermediary can become a cash-flow and patient-service crisis across an ecosystem.

SOURCE: U.S. Securities and Exchange Commission • UnitedHealth Group Form 8-K, 21 February 2024 • sec.gov • CLICK TO OPEN OFFICIAL SOURCE

Source: https://www.sec.gov/Archives/edgar/data/731766/000073176624000045/unh-20240221.htm

Decision room: Change Healthcare cyberattack

TEAM DISCUSSION • 8 MIN

1

TRIAGE

Which evidence would you trust first: Service interruption across connected customers and loss of access to transactional networks?

2

DECISION

Which immediate action limits electronic claims, payments and pharmacy transactions were disrupted across healthcare providers without destroying evidence?

3

CONTROL

Which one control should be verified first from: Strong remote-access identity, supplier concentration analysis, alternate clearing routes, isolation plans and recovery communications?

4

TRANSFER

Where could the same entry path exist in a personal account or small business today?

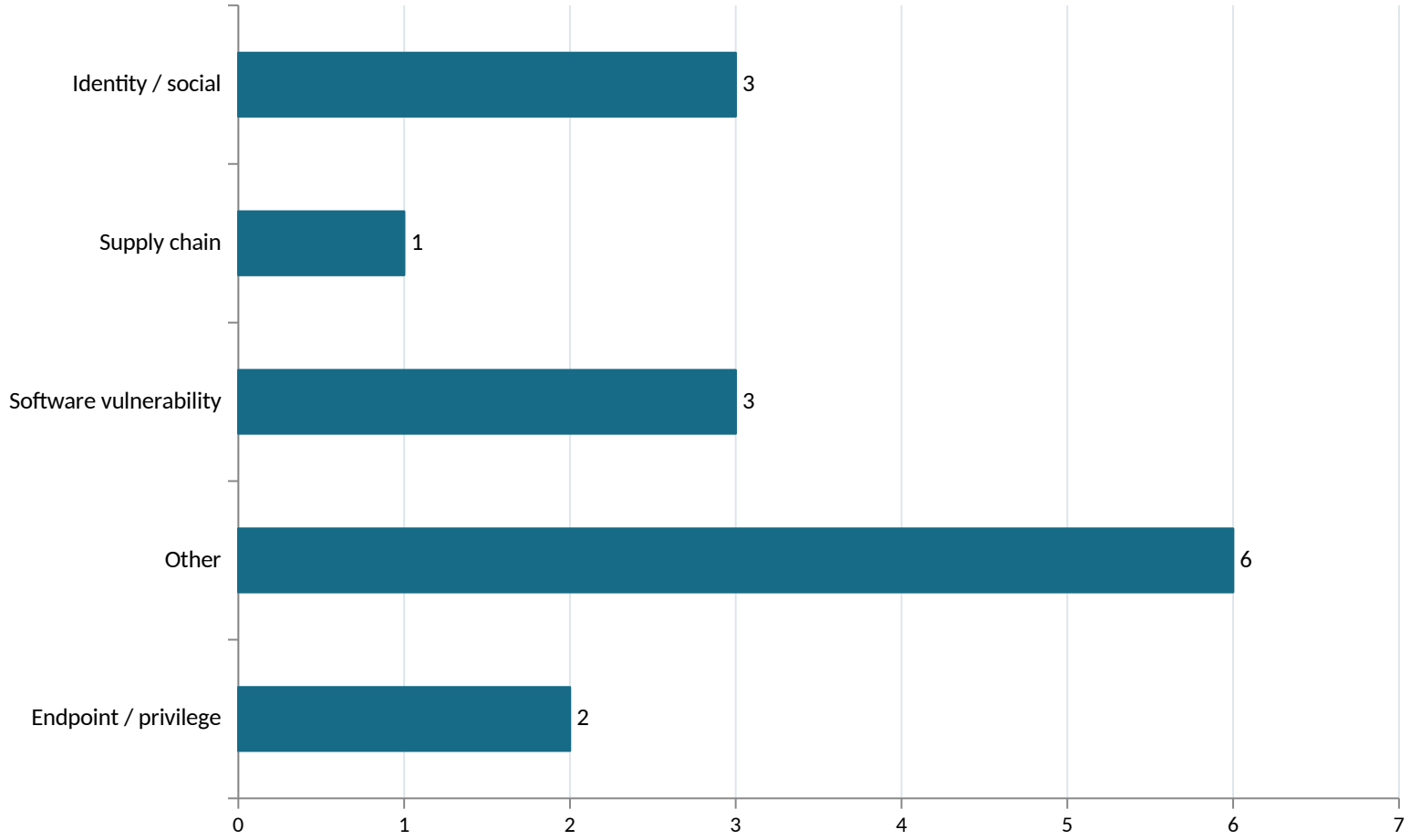


SOURCE: U.S. Securities and Exchange Commission • UnitedHealth Group Form 8-K, 21 February 2024 • sec.gov • CLICK TO OPEN OFFICIAL SOURCE

Source: https://www.sec.gov/Archives/edgar/data/731766/000073176624000045/unh-20240221.htm

EDITABLE DATA VIEW

How the course case set was entered



READ THE SIGNAL

This chart describes only the 15 teaching cases in this course. It is a pattern-recognition aid, not an estimate of attack prevalence.

Chart is native and editable in PowerPoint.

Source: Course case analysis; see case-study source register.

TRANSFER THE LESSON — NOT THE HEADLINE

Case patterns that repeat

Known weakness

A public issue remained reachable or unverified after remediation.

Identity gap

Dormant, over-privileged or weakly verified accounts carried trust.

Supplier path

One dependency multiplied exposure across many organisations.

Detection delay

Signals existed but were fragmented, noisy or unowned.

Business process

Payment, recovery or support workflows bypassed independent verification.

Recovery readiness

Operations depended on inventories, backups and practiced decisions.

PRACTICE LAB 01

Threat Triage with the Cybersecurity Simulator

25 minutes | LO1 | Cybersecurity Simulator

BUSINESS SCENARIO

A small Singapore design agency receives five mixed security alerts. Decide what is benign, suspicious or urgent.

ACCEPTED EVIDENCE

Completed triage table • Severity rationale • Screenshot of highest-risk decision • One escalation statement

DECISION

What is the risk and how certain are you?

SAFE ACTION

What is the least harmful next action?

ESCALATION

Who needs what evidence, by when?

Detailed procedures, mock data and verification criteria are in the Learner Guide and the lab folder.

Source: https://alfredang.github.io/cybersecuritysimulator/

PRACTICE LAB 02

Detect Phishing and Business Email Compromise

35 minutes

LO1, LO3

Cybersecurity Simulator + Regex Generator

BUSINESS SCENARIO

Accounts staff must detect a lookalike supplier domain before approving a high-value transfer.

ACCEPTED EVIDENCE

Annotated inbox • Tested regex • Detection test results • Independent verification script • Escalation decision

DECISION

What is the risk and how certain are you?

SAFE ACTION

What is the least harmful next action?

ESCALATION

Who needs what evidence, by when?

Detailed procedures, mock data and verification criteria are in the Learner Guide and the lab folder.

Source: https://alfredang.github.io/regexgenerator/

PRACTICE LAB 03

Recognise Malware and Endpoint Warning Signs

30 minutes

LO1, LO3

Cybersecurity Simulator

BUSINESS SCENARIO

Endpoint alerts reveal suspicious processes, persistence and outbound connections on an employee laptop.

ACCEPTED EVIDENCE

Malware classification • Attack-chain map • Evidence vs inference notes • Containment decision • Escalation note

DECISION

What is the risk and how certain are you?

SAFE ACTION

What is the least harmful next action?

ESCALATION

Who needs what evidence, by when?

Detailed procedures, mock data and verification criteria are in the Learner Guide and the lab folder.

Source: https://alfredang.github.io/cybersecuritysimulator/

02

TOPIC

Protect personal information

Build a practical baseline for identity, devices, data, online activity and recovery.

SEE

DECIDE

ACT

LAYERED SAFEGUARDS

The personal protection baseline

Identity

Unique long passphrases, password manager, MFA/passkeys, disable dormant accounts

Devices

Automatic updates, supported software, standard-user access, encryption, screen lock

Data

Classification, least access, secure transfer, retention limits, tested backups

Networks

Secure Wi-Fi, segmentation, firewalling, VPN where appropriate, monitored remote access

People & process

Verify unusual requests, dual approval, reporting channels, rehearsed escalation

Modern password practice

USE

- Unique long passphrases
- A reputable password manager
- MFA or passkeys
- Change credentials after suspected compromise

AVOID

- Reusing one password
- Predictable substitutions
- Routine forced changes without cause
- Sharing passwords in messages or documents

Source: https://www.microsoft.com/en-us/security/business/security-101/what-is-cybersecurity

MFA is stronger when factors resist phishing

Something you know

Password or PIN; vulnerable to guessing, theft and reuse.

Something you have

Authenticator, security key or registered device.

Something you are

Biometric used locally to unlock a trusted authenticator.

Phishing resistance

Passkeys and security keys bind authentication to the legitimate service.

Secure a device through its lifecycle

1 Acquire

Supported hardware and trusted software sources

2 Configure

Encryption, standard user, screen lock, least permissions

3 Maintain

Automatic updates, health checks and safe applications

4 Monitor

Alerts, account activity and unusual behaviour

5 Retire

Backup, sign out, wipe and verify data removal

The Learner Guide contains platform-neutral verification steps.

3–2–1 + VERIFY

Backups are a recovery control

3 copies

Keep production data plus two backup copies.

2 media or locations

Avoid one failure affecting every copy.

1 offline or immutable

Make one copy resistant to ransomware and account takeover.

Test restore

A backup is only useful when the right data can be restored in time.

Safer browsing and online transactions

Navigate deliberately

Use bookmarks or type a known address for important services.

Check the destination

HTTPS protects the connection; it does not prove the business is honest.

Limit payment exposure

Use trusted payment channels, alerts and transaction limits.

Reduce data

Share only what the transaction genuinely needs.

Pause on change

New bank details, delivery fees or recovery prompts require verification.

Report fast

Contact the bank and relevant authorities quickly when fraud is suspected.

Home and public-network safeguards

Home router

Change default admin credentials, update firmware, use strong Wi-Fi encryption and disable unused remote admin.

Guest separation

Keep visitors and smart devices away from trusted work devices where supported.

Public Wi-Fi

Prefer mobile data for sensitive work; verify the network and avoid unexpected captive portals.

VPN

Protects traffic to the VPN endpoint; it does not make unsafe sites or compromised devices trustworthy.

When an account may be compromised

1 Stop

Do not approve unfamiliar prompts or continue risky activity

2 Use a clean path

Access the service from a trusted device and known address

3 Secure

Change the password, revoke sessions and strengthen MFA

4 Review

Recovery details, forwarding rules, connected apps and transactions

5 Report

Provider, employer, bank or authority based on impact

Speed matters, but evidence and safe access matter too.

PRACTICE LAB 06

Protect Data with Hashing and Encryption

30 minutes

LO2, LO3

Cryptography Toolkit

BUSINESS SCENARIO

A clinic must distinguish confidentiality, integrity checking and password storage when protecting mock records.

ACCEPTED EVIDENCE

Data classification • Hash comparison • Encrypted/decrypted sample • Key-handling checklist • Security explanation

DECISION

What is the risk and how certain are you?

SAFE ACTION

What is the least harmful next action?

ESCALATION

Who needs what evidence, by when?

Detailed procedures, mock data and verification criteria are in the Learner Guide and the lab folder.

Source: https://alfredang.github.io/cryptography-toolkit/

03

TOPIC

Protect business information

Prioritise vulnerabilities, layer controls, monitor signals and escalate incidents.

SEE

DECIDE

ACT

AREAS PRONE TO ATTACK

Vulnerabilities are more than software bugs

Unpatched software

Known flaw remains exploitable

Weak or reused credentials

One stolen secret unlocks several services

Excess privilege

Compromise reaches more data and systems

Misconfiguration

Security feature exists but is unsafe

Internet exposure

Attack surface is reachable by anyone

Unsafe input handling

Application trusts hostile input

Third-party dependency

Supplier weakness becomes customer exposure

Human process gap

Urgency or authority bypasses verification

Source: https://www.ncsc.gov.uk/collection/vulnerability-management/understanding-vulnerabilities

Unpatched software: exposure, treatment and proof



1 WHY IT IS EXPLOITABLE

Known flaw remains exploitable

2 RISK TREATMENT

Asset inventory + risk-based patch SLA

3 VERIFY THE CONTROL

Inventory shows the affected version is absent and a rescan confirms closure.

A vulnerability is not closed until the fix is verified on the correct asset and regressions are monitored.

Source: https://www.ncsc.gov.uk/collection/vulnerability-management/understanding-vulnerabilities

Weak or reused credentials: exposure, treatment and proof

1 WHY IT IS EXPLOITABLE

One stolen secret unlocks several services

2 RISK TREATMENT

Password manager + MFA/passkeys + breach response

3 VERIFY THE CONTROL

MFA coverage, breach response and session review are tested for priority accounts.

A vulnerability is not closed until the fix is verified on the correct asset and regressions are monitored.



Source: https://www.ncsc.gov.uk/collection/vulnerability-management/understanding-vulnerabilities

Excess privilege: exposure, treatment and proof



1 WHY IT IS EXPLOITABLE

Compromise reaches more data and systems

2 RISK TREATMENT

Least privilege + access reviews

3 VERIFY THE CONTROL

Access review confirms only current roles retain approved permissions.

A vulnerability is not closed until the fix is verified on the correct asset and regressions are monitored.

Source: https://www.ncsc.gov.uk/collection/vulnerability-management/understanding-vulnerabilities

Misconfiguration: exposure, treatment and proof

1 WHY IT IS EXPLOITABLE

Security feature exists but is unsafe

2 RISK TREATMENT

Secure baseline + continuous configuration checks

3 VERIFY THE CONTROL

The secure baseline is measured continuously and exceptions have owners and expiry dates.

A vulnerability is not closed until the fix is verified on the correct asset and regressions are monitored.



Source: https://www.ncsc.gov.uk/collection/vulnerability-management/understanding-vulnerabilities

Internet exposure: exposure, treatment and proof



1 WHY IT IS EXPLOITABLE

Attack surface is reachable by anyone

2 RISK TREATMENT

Minimise services + strong authentication + monitoring

3 VERIFY THE CONTROL

External discovery confirms only required services are reachable and monitored.

A vulnerability is not closed until the fix is verified on the correct asset and regressions are monitored.

Source: https://www.ncsc.gov.uk/collection/vulnerability-management/understanding-vulnerabilities

Unsafe input handling: exposure, treatment and proof

1 WHY IT IS EXPLOITABLE

Application trusts hostile input

2 RISK TREATMENT

Validation + parameterised queries + testing

3 VERIFY THE CONTROL

Security tests demonstrate hostile input cannot change the intended operation.

A vulnerability is not closed until the fix is verified on the correct asset and regressions are monitored.



Source: https://www.ncsc.gov.uk/collection/vulnerability-management/understanding-vulnerabilities

Third-party dependency: exposure, treatment and proof



1 WHY IT IS EXPLOITABLE

Supplier weakness becomes customer exposure

2 RISK TREATMENT

Supplier inventory + contract controls + monitoring

3 VERIFY THE CONTROL

The supplier inventory maps data, access, concentration, assurance and exit options.

A vulnerability is not closed until the fix is verified on the correct asset and regressions are monitored.

Source: https://www.ncsc.gov.uk/collection/vulnerability-management/understanding-vulnerabilities

Human process gap: exposure, treatment and proof

1 WHY IT IS EXPLOITABLE

Urgency or authority bypasses verification

2 RISK TREATMENT

Independent verification + dual approval + rehearsal

3 VERIFY THE CONTROL

A rehearsal proves unusual payments, recovery and access changes trigger independent checks.

A vulnerability is not closed until the fix is verified on the correct asset and regressions are monitored.



Source: https://www.ncsc.gov.uk/collection/vulnerability-management/understanding-vulnerabilities

DO NOT CONFUSE THE TERMS

Vulnerability, exploit and threat

1 **Vulnerability**

A weakness, insecure feature or user/process error

2 **Exploit**

Code or technique that uses the weakness

3 **Threat**

Actor or event with capability and intent

4 **Exposure**

Conditions that make exploitation reachable

5 **Risk**

Likelihood and business consequence after controls

Source: https://www.ncsc.gov.uk/collection/vulnerability-management/understanding-vulnerabilities

Common vulnerable areas and treatments

Unpatched software

Asset inventory + risk-based patch SLA

Weak or reused credentials

Password manager + MFA/passkeys + breach response

Excess privilege

Least privilege + access reviews

Misconfiguration

Secure baseline + continuous configuration checks

Internet exposure

Minimise services + strong authentication + monitoring

Unsafe input handling

Validation + parameterised queries + testing

Third-party dependency

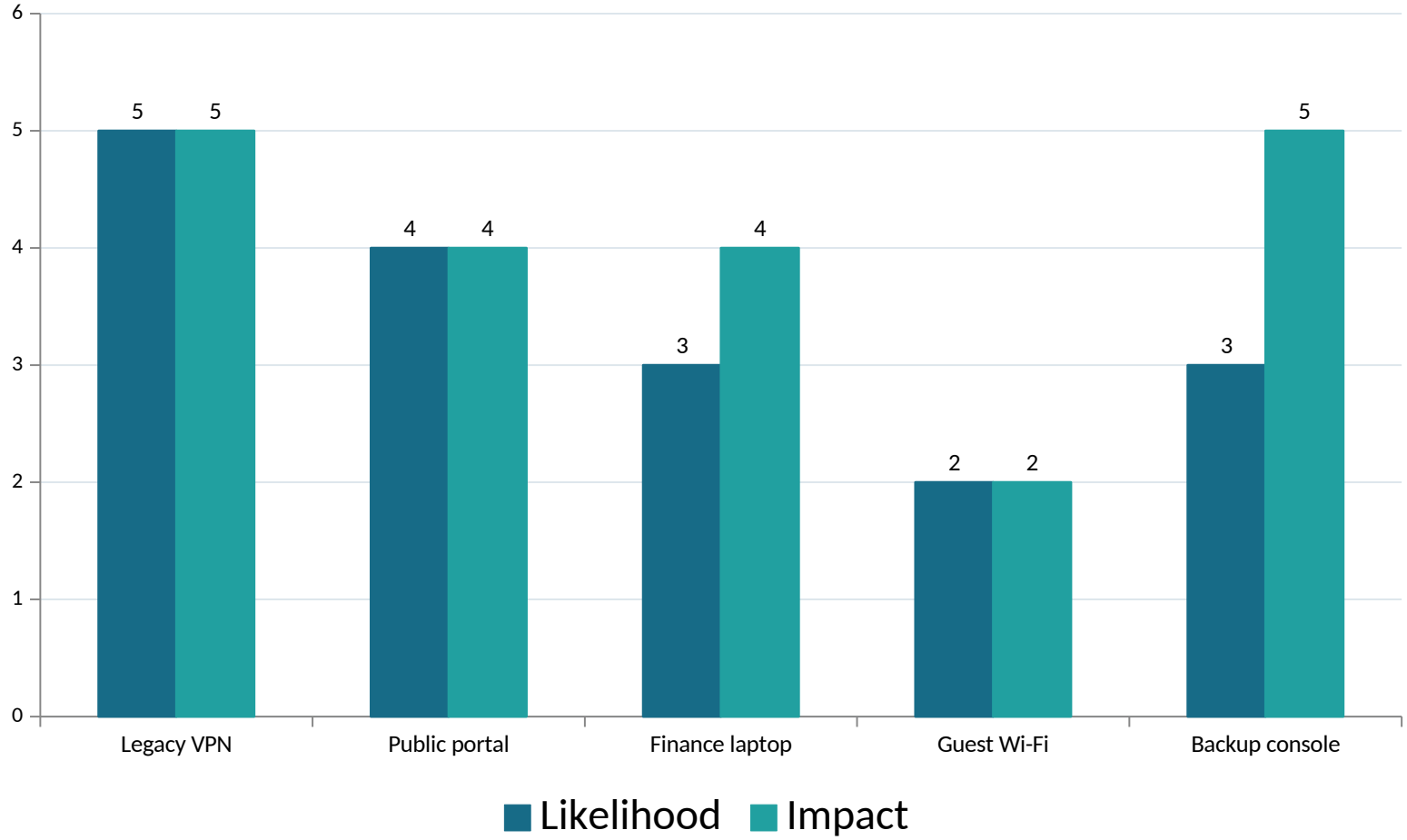
Supplier inventory + contract controls + monitoring

Human process gap

Independent verification + dual approval + rehearsal

EDITABLE DATA VIEW

Prioritise by business context



READ THE SIGNAL

The same technical severity can produce different business priorities. Reachability, data sensitivity, exploit activity and control strength change the decision.

Chart is native and editable in PowerPoint.

Source: Synthetic Lab 04 risk register.

A complete vulnerability-management loop

1 Discover

Inventory assets, services, software and owners

2 Assess

Validate exposure, exploitability and business impact

3 Treat

Patch, isolate, configure, compensate, accept or retire

4 Verify

Confirm the fix and detect regressions

5 Learn

Improve ownership, standards and response time

Closing a ticket is not the same as closing the exposure.

Source: https://www.ncsc.gov.uk/collection/vulnerability-management/understanding-vulnerabilities

Web application risk in plain language

Injection

Hostile input changes a command or query. Use parameterised operations and strict validation.

Broken access control

A user can reach data or actions beyond their role. Enforce checks on the server.

Authentication failure

Sessions, passwords or recovery are weak. Use secure identity design and monitoring.

Security misconfiguration

Defaults, debug features or permissions expose capability. Use hardened baselines.

Source: https://owasp.org/www-community/attacks/SQL_Injection

IDS, IPS and endpoint detection

DETECT

- IDS observes and alerts on network patterns
- Endpoint detection correlates process, identity and file behaviour
- Logs need owners, context and retention
- Honeypots can reveal unexpected activity

PREVENT / CONTAIN

- IPS can block matched traffic
- Application control limits execution
- Segmentation limits movement
- Automated action needs safe thresholds and review

AUTHORISED TESTING ONLY

Penetration testing needs governance

Written authority

Named owner, target, dates and permitted techniques.

Rules of engagement

Safety limits, data handling, stop conditions and contacts.

Evidence discipline

Minimum necessary data, redaction and secure retention.

Remediation

Business impact, owner, due date, verification and retest.

Governance turns controls into routine

Ownership

Every asset, risk, alert and action has an accountable role.

Policy

Clear requirements for access, data, suppliers, change, backup and reporting.

Metrics

Patch time, MFA coverage, restore success, alert response and overdue risk.

Assurance

Review, test and independently verify critical controls.

Training

Role-specific rehearsal, not one-time awareness.

Improvement

Incidents and near misses change design and behaviour.

Business continuity and cyber recovery



Recovery is a business decision supported by technology.

Escalate with structure

1 Situation

What is happening now?

2 Evidence

What facts and indicators support it?

3 Impact

Which people, data, services or money are at risk?

4 Action

What has been contained or preserved?

5 Decision

What authority, resource or communication is needed?

State uncertainty explicitly; never present inference as confirmed fact.

When to escalate immediately

Safety or essential service

Potential harm to people or critical operations.

Sensitive data

Possible loss of personal, health, financial or confidential data.

Privilege

Administrator, finance, executive or security-control account involved.

Active spread

Multiple systems, lateral movement or command-and-control signs.

Money movement

Payment change, unauthorised transfer or extortion demand.

Uncertainty with high impact

Evidence is incomplete but delay could materially worsen outcomes.

PRACTICE LAB 04

Prioritise Vulnerabilities by Business Risk

35 minutes

LO1

Ethical Hacking Simulator

BUSINESS SCENARIO

An SME has limited maintenance time and must decide which exposures to address first.

ACCEPTED EVIDENCE

Ranked risk register • Top-three rationale • Treatment decisions • Owners and due dates • Verification plan

DECISION

What is the risk and how certain are you?

SAFE ACTION

What is the least harmful next action?

ESCALATION

Who needs what evidence, by when?

Detailed procedures, mock data and verification criteria are in the Learner Guide and the lab folder.

Source: https://alfredang.github.io/ethnicalhacking/

PRACTICE LAB 05

Authorised Web-Risk Review in FauxBank

40 minutes | LO1, LO3 | FauxBank

BUSINESS SCENARIO

A bank training application needs a safe, evidence-based review of authentication and input-handling risks.

ACCEPTED EVIDENCE

Signed scope check • Completed test plan • Redacted screenshots • Risk statements • Responsible report

DECISION

What is the risk and how certain are you?

SAFE ACTION

What is the least harmful next action?

ESCALATION

Who needs what evidence, by when?

Detailed procedures, mock data and verification criteria are in the Learner Guide and the lab folder.

Source: https://pentest-fauxbank.vercel.app/

PRACTICE LAB 07

Map Network Exposure and Segmentation

35 minutes

LO1, LO3

IP Calculator

BUSINESS SCENARIO

A retail office needs separate zones for staff, guests, point-of-sale devices and management systems.

ACCEPTED EVIDENCE

Subnet plan • Asset-to-zone mapping • Allowed-flow matrix • Exposure decisions • Residual risk

DECISION

What is the risk and how certain are you?

SAFE ACTION

What is the least harmful next action?

ESCALATION

Who needs what evidence, by when?

Detailed procedures, mock data and verification criteria are in the Learner Guide and the lab folder.

Source: https://alfredang.github.io/ipcalculator/

PRACTICE LAB 08

Triage Suspicious Network Traffic

35 minutes

LO1, LO3

PCAP Analyzer

BUSINESS SCENARIO

A workstation generates unusual DNS and outbound traffic after a user opens an attachment.

ACCEPTED EVIDENCE

Traffic findings • Five-event timeline • Earliest reliable signal • Containment decision • Handover note

DECISION

What is the risk and how certain are you?

SAFE ACTION

What is the least harmful next action?

ESCALATION

Who needs what evidence, by when?

Detailed procedures, mock data and verification criteria are in the Learner Guide and the lab folder.

Source: https://alfredang.github.io/pcapanalyzer/

PRACTICE LAB 09

Design and Test a Detection Rule

30 minutes | LO3 | Regex Generator

BUSINESS SCENARIO

A security team needs a transparent rule for repeated failed logins and suspicious usernames.

ACCEPTED EVIDENCE

Tested regex • Confusion matrix • Threshold decision • Rule documentation • Escalation criteria

DECISION

What is the risk and how certain are you?

SAFE ACTION

What is the least harmful next action?

ESCALATION

Who needs what evidence, by when?

Detailed procedures, mock data and verification criteria are in the Learner Guide and the lab folder.

Source: https://alfredang.github.io/regexgenerator/

PRACTICE LAB 10

Integrated Incident Response and Escalation

45 minutes

LO1–LO3

Integrated toolkit

BUSINESS SCENARIO

A supplier-payment email, endpoint alert and suspicious network connection may form one business-impacting incident.

ACCEPTED EVIDENCE

Correlated timeline • CIA and business-impact assessment • Decision log • Escalation record • Three-minute briefing • After-action improvement

DECISION

What is the risk and how certain are you?

SAFE ACTION

What is the least harmful next action?

ESCALATION

Who needs what evidence, by when?

Detailed procedures, mock data and verification criteria are in the Learner Guide and the lab folder.

Source: https://alfredang.github.io/cybersecuritysimulator/

Your 30-day cybersecurity action plan

Within 24 hours

Enable MFA on priority accounts; verify reporting contacts; test one backup restore.

Within 7 days

Remove dormant access; update exposed systems; document payment-change verification.

Within 30 days

Run a tabletop incident; review supplier exposure; close the top three risks with evidence.

Key takeaways

See the chain

Assets, threats, vulnerabilities and business processes combine to create risk.

Layer safeguards

Identity, devices, data, networks, people and recovery support one another.

Verify controls

A configured control is not proven until it is tested and monitored.

Escalate early

Structured evidence and independent verification protect outcomes.

AUTHORITATIVE AND APPROVED REFERENCES

Source register

IBM

Cybersecurity overview

Cisco

What is cybersecurity?

Microsoft

What is cybersecurity?

Kaspersky

Cybersecurity definition

Fortinet

Cybersecurity glossary

Check Point

What is cybersecurity?

CSA Singapore

Cybersecurity guidance

NCSC

Understanding vulnerabilities

OWASP

Vulnerabilities and SQL injection

REAL-WORLD EVIDENCE

Case-study source register — Part 1

SingHealth data breach

Ministry of Health Singapore • SingHealth's IT System Target of Cyberattack

WannaCry and the NHS

UK National Audit Office • Investigation: WannaCry Cyber Attack and the NHS

Equifax breach

U.S. House Committee on Oversight • The Equifax Data Breach

SolarWinds supply-chain compromise

CISA • Active Exploitation of SolarWinds Software

Colonial Pipeline

CISA • IAM Case Study: Colonial Pipeline

Log4Shell exploitation

CISA • Mitigating Log4Shell and Log4j Vulnerabilities

MOVEit mass exploitation

CISA • CL0P Exploits MOVEit Vulnerability

MGM Resorts cyber incident

U.S. Securities and Exchange Commission • MGM Resorts International Form 8-K

Case-study source register — Part 2

Singapore commodity-trader BEC

Singapore Police Force • Recovery of Over USD 6.6 Million in BEC Case

Target payment-card breach

U.S. Senate Committee on Commerce • Target's Missed Opportunities to Stop a Data Breach

NotPetya disruption at Maersk

UK Department for Transport • Cyber Security for Ports and Port Systems

Marriott Starwood data breach

UK Information Commissioner's Office • Marriott International Monetary Penalty Notice

Capital One cloud data breach

U.S. Department of Justice • Seattle Tech Worker Arrested for Data Theft

Norsk Hydro LockerGoga ransomware

Dragos Industrial Cybersecurity Centre • What Is OT Malware?

Change Healthcare cyberattack

U.S. Securities and Exchange Commission • UnitedHealth Group Form 8-K, 21 February 2024

Thank you

Recognise

Pause, inspect context and connect signals.

Reduce

Apply the smallest effective control, then verify it.

Escalate

Share facts, impact, action and the decision required.